

KMOU Division of Marine System Engineering

Introduction to Internal Combustion Engine



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Lesson content for each week

Ch0_Introduction to the Class

Ch6_Piston Fittings and Crankshaft

Ch1_Overview of the Internal Combustion Engine

Ch7_Lube Oil System

Ch2_Thermodynamics and Compressed Air System

Ch8_Cooling Water System

Ch3_Marine Fuels and Internal Fuel Oil System

Ch9_Introduction to the Main Engine

Ch4_Intake and Exhaust System

Ch10_Alternative Fuels and Alternative Fuel Systems

Ch5_Turbocharger

Ch11_Environmental Regulations and Exhaust Gas Aftertreatment Systems

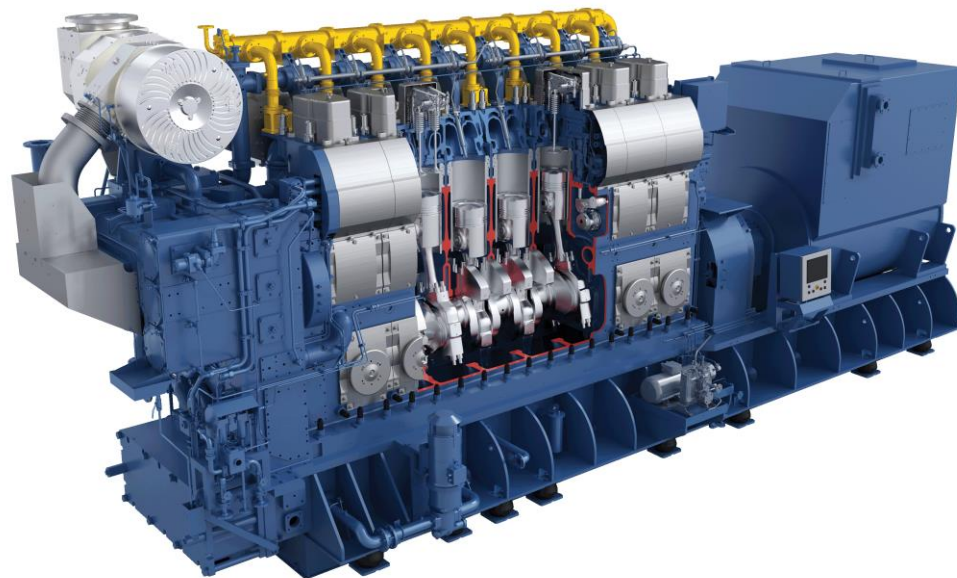
Midterm

Ch12_Tools, Apprentice Engineer Duties, and Career Direction

Final Exam

Engine type identification for MAN B&W

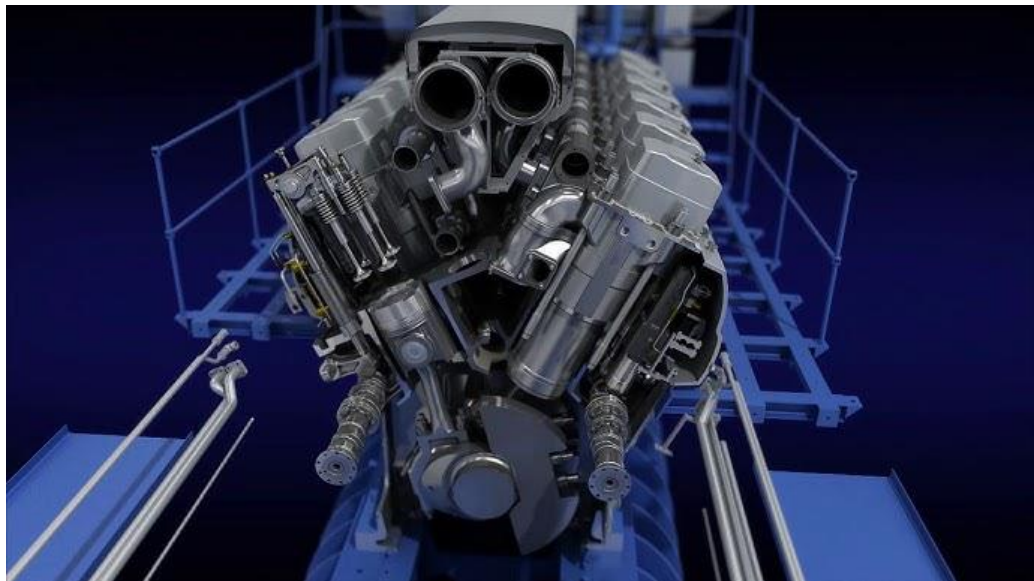
	6	L	28/32	H	MCR
No of cylinders					
5, 6, 7, 8, 9 12, 16, 18					
Engine Type					
L : In-line V : V-built					
Cyl. diam/stroke					
16/24 : 160/240 21/31 : 210/310 23/30 : 225/300 27/38 : 270/380 28/32 : 280/320 32/40 : 320/400					
Design Variant					
Rating					
MCR : Maximum continuous rating ECR : Economy continuous rating					



G/E (In-line type) with 8 cylinders

Engine type identification for MAN B&W

	6	L	28/32	H	MCR
No of cylinders					
5, 6, 7, 8, 9 12, 16, 18					
Engine Type					
L : In-line V : V-built					
Cyl. diam/stroke					
16/24 : 160/240 21/31 : 210/310 23/30 : 225/300 27/38 : 270/380 28/32 : 280/320 32/40 : 320/400					
Design Variant					
Rating					
MCR : Maximum continuous rating ECR : Economy continuous rating					



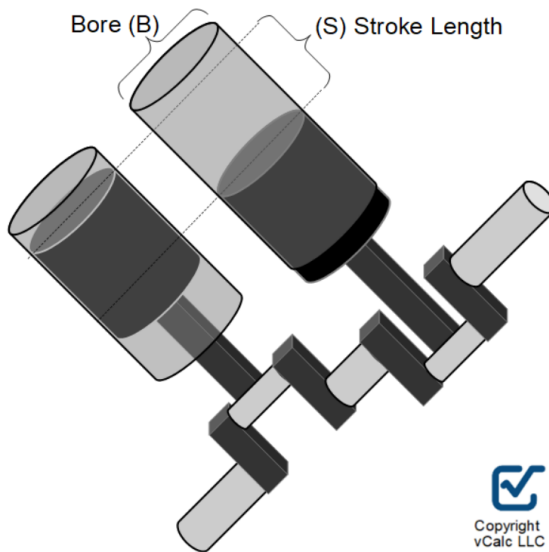
G/E (V-built type)

Engine type identification for MAN B&W

	6	L	28/32	H	MCR
No of cylinders					
5, 6, 7, 8, 9 12, 16, 18					
Engine Type					
L : In-line V : V-built					
Cyl. diam/stroke					
16/24 : 160/240 21/31 : 210/310 23/30 : 225/300 27/38 : 270/380 28/32 : 280/320 32/40 : 320/400					
Design Variant					
Rating					

Bore : 280 mm

Stroke : 320 mm



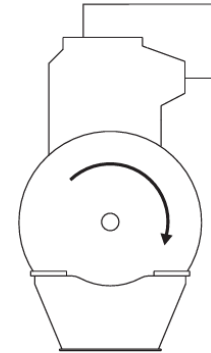
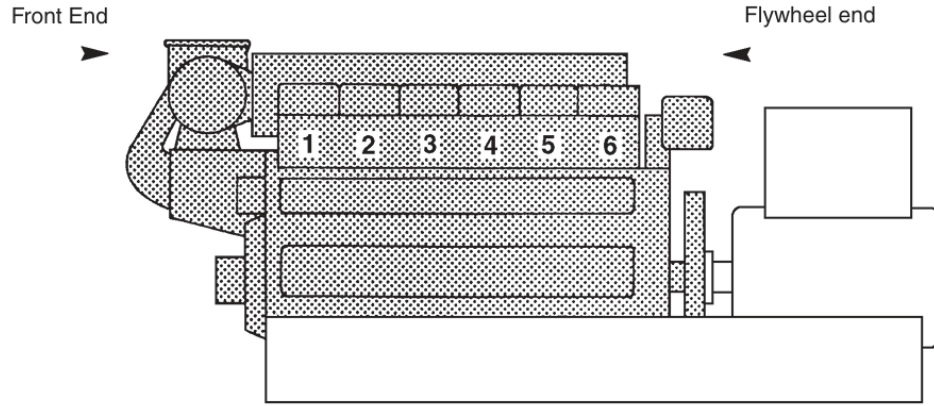
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MCR : Maximum continuous rating
ECR : Economy continuous rating

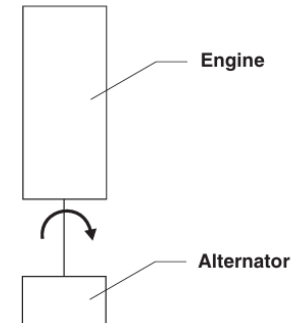
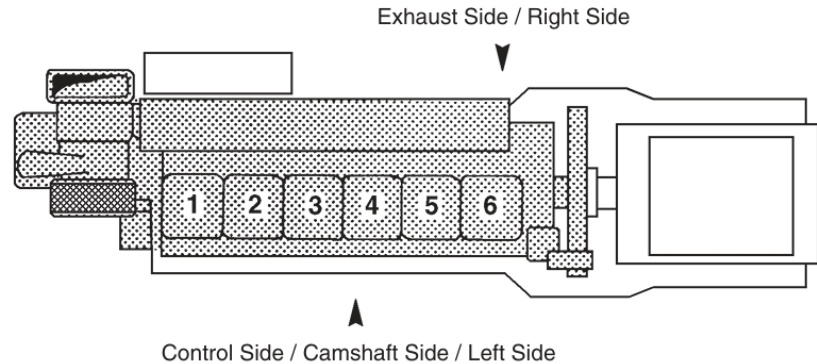


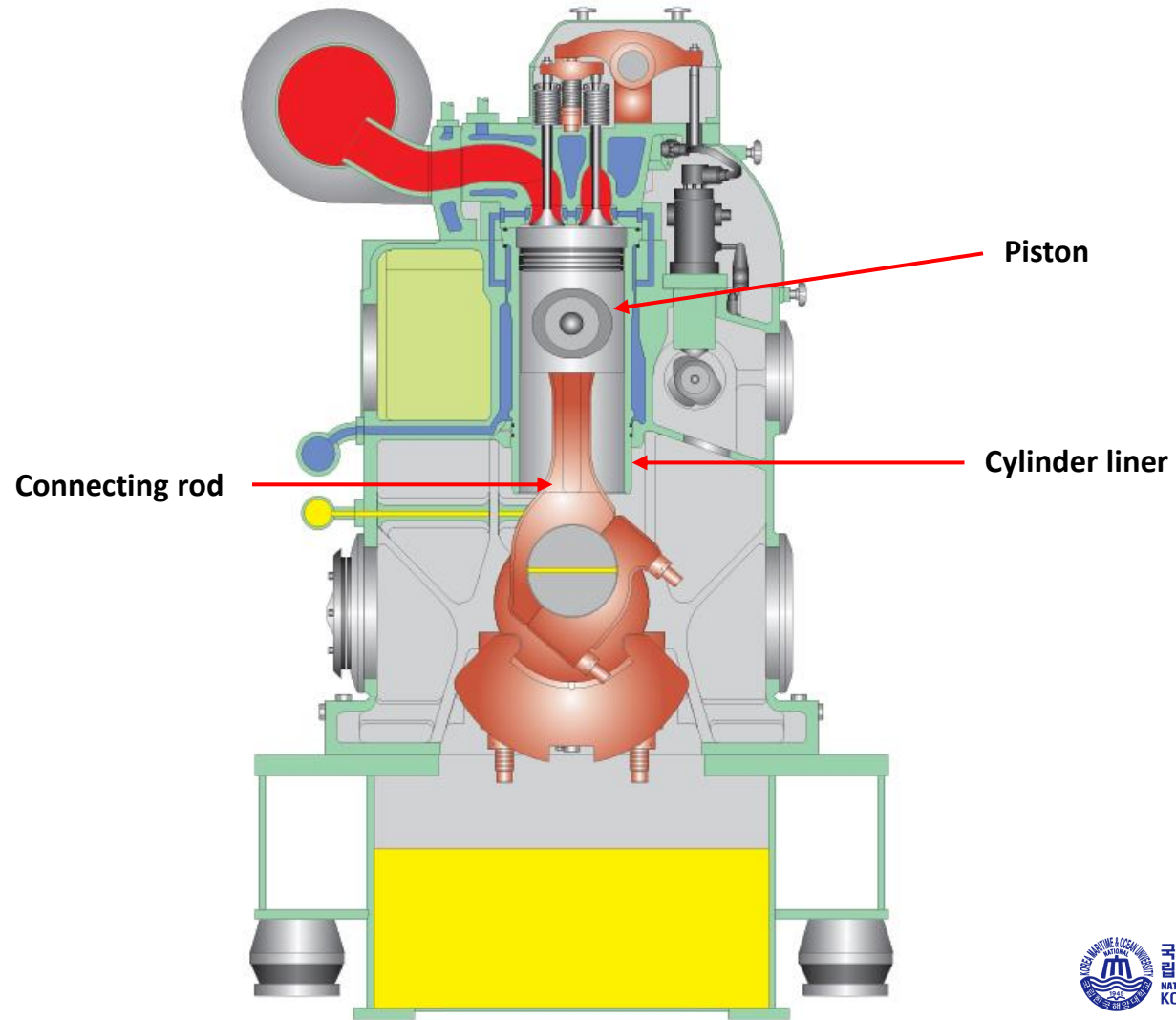
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Engine type identification for MAN B&W



Direction of rotation seen from flywheel end "Clockwise"

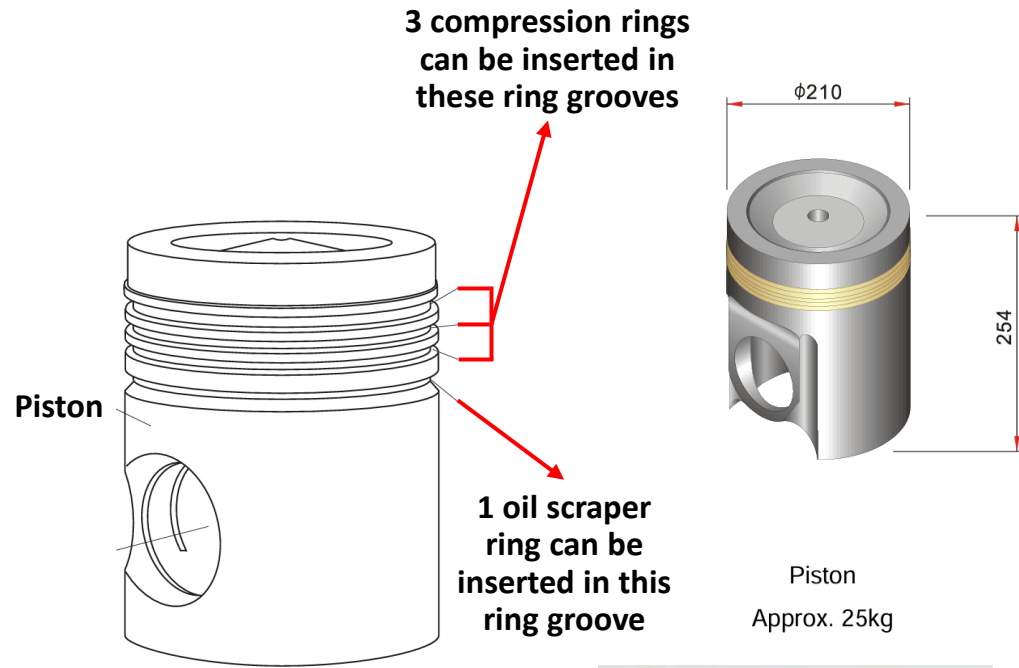




Piston

The piston, which is oil-cooled and of the monobloc type made of nodular cast iron, is equipped with 3 compression rings and 1 scraper ring.

By use of a combination of compression rings with different barrel-shaped profiles and chrome-plated running surface on all rings, the piston ring pack is optimized for maximum sealing effect and minimum wear rate.



1 compression ring is being inserted by special ring opener

2 compression rings can be inserted in these ring grooves

1 oil scraper ring was inserted in the ring groove



The piston has to be scrapped if:

A) The wear limit on the testing mandrel is exceeded, see fig 1A

or

B) The clearance between the new piston/scrapper ring and ring groove is exceeded, see fig 1B.

Note: At each piston overhaul:

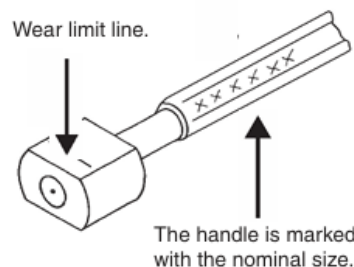
- **The piston and scraper ring must be exchanged.**
- **The cylinder liner must be honed according to the instructions.**

	<u>Piston and oil scraper ring.</u> Nominal size.	<u>New ring grooves.</u> Tolerances.	<u>Ring grooves.</u> Max. wear limit.
Piston ring no 1	New 5.0 mm	5.0 mm +0.18 +0.16	5.43 mm
Piston ring no 2	New 5.0 mm	5.0 mm +0.14 +0.12	5.43 mm
Piston ring no 3	New 4.95 mm	5.0 mm +0.14 +0.12	5.43 mm
Scraper ring	New 8.0 mm	8.0 mm +0.12 +0.10	8.43 mm

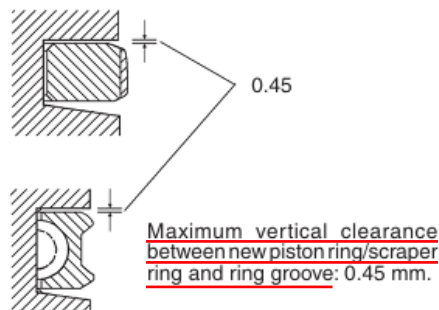
Table 1 Nominal size, new ring groove tolerance and wear limit for ring grooves

A) Testing Mandrel for Ring Grooves

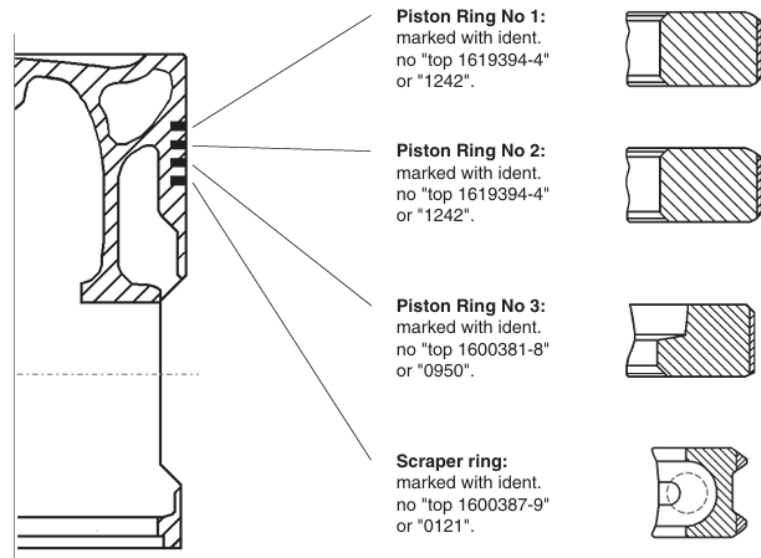
If the wear limit (2 mm mark) on the testing mandrel is exceeded, the specified max. wear limits are exceeded, and the piston must be scrapped.



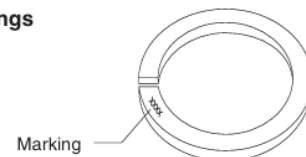
B) Clearance Ring/Groove



Position of Piston and Scraper Rings



Marking of Piston and Scraper Rings



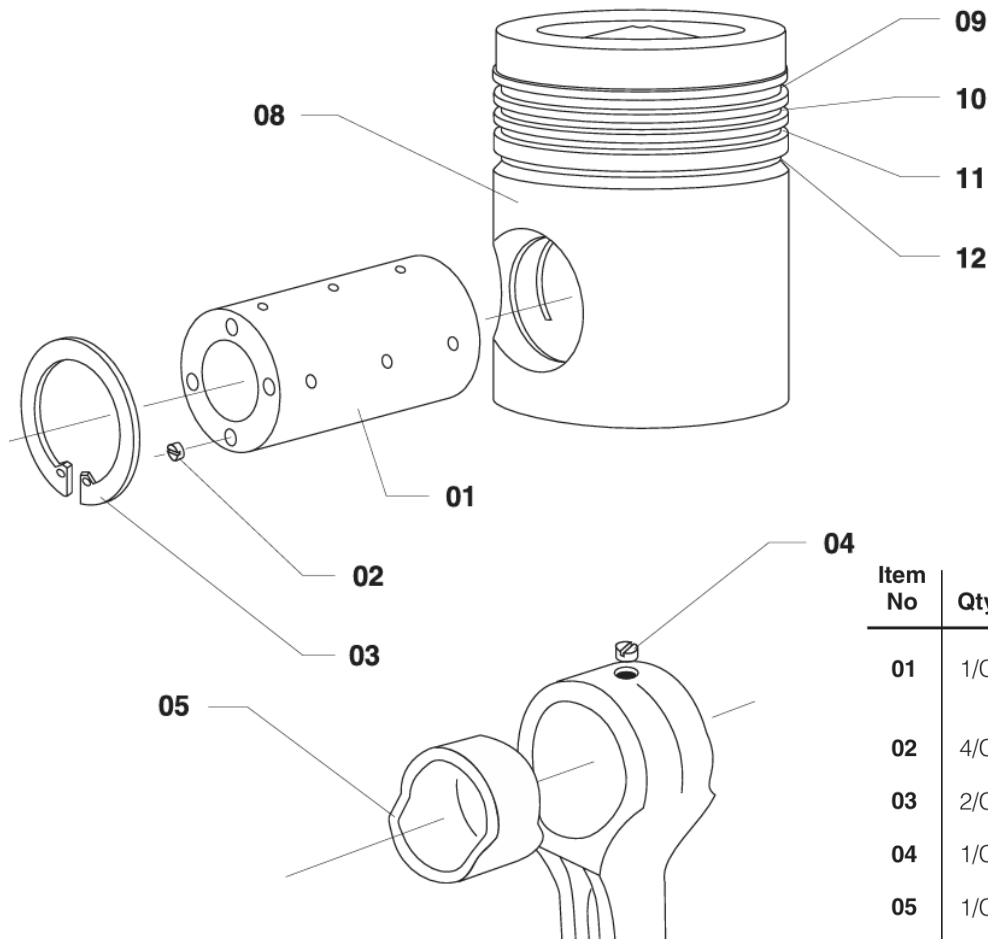
Identification marks to face upwards against the piston crown when mounted.

Note: The marking may include other figures than mentioned above, for instance trade mark and production codes.

Piston pin

The piston pin is fully floating which means that it can turn freely in the pin bosses of the piston as well as in the connecting rod bush.

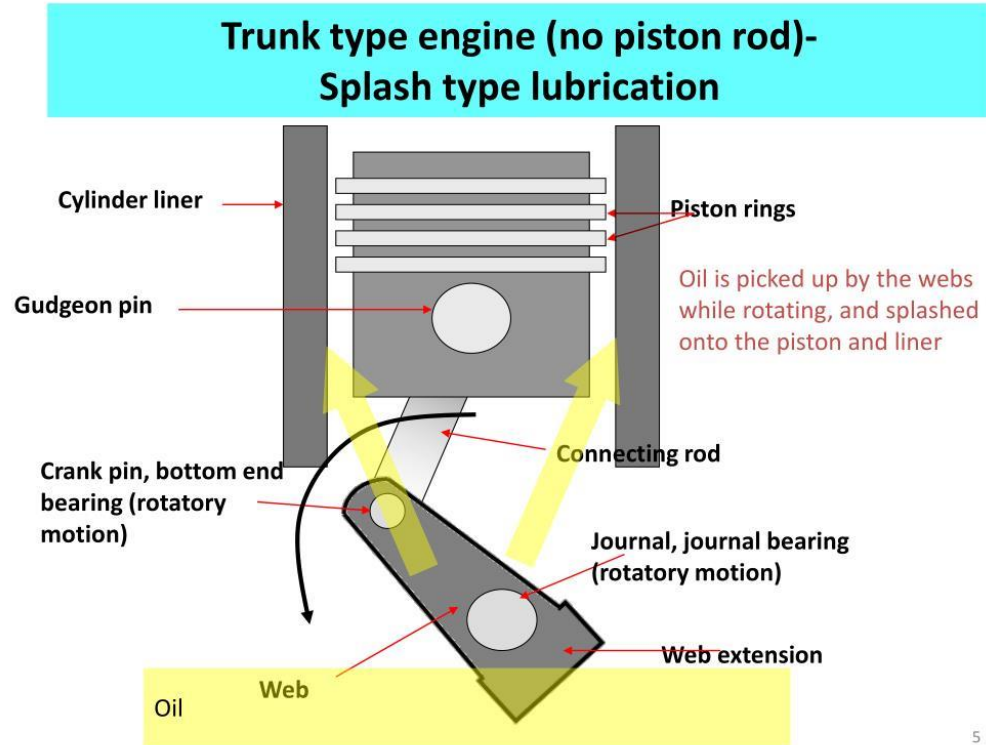
The piston pin is prevented from moving in the axial direction by two circlips (retainer rings).



Item No	Qty.	Designation
01	1/C	Piston pin incl. item No 02
02	4/C	Socket screw
03	2/C	Retaining ring
04	1/C	Plug screw
05	1/C	Bush for connecting rod

Lubrication between piston and cylinder liner

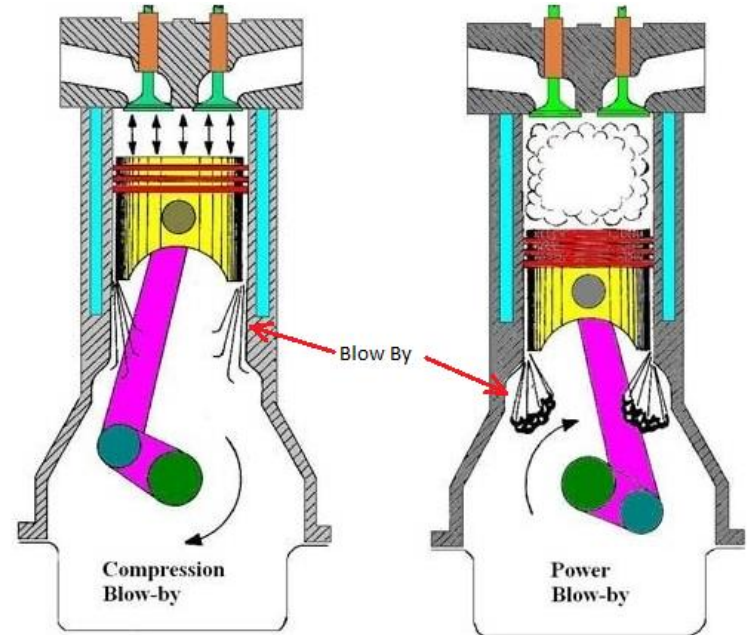
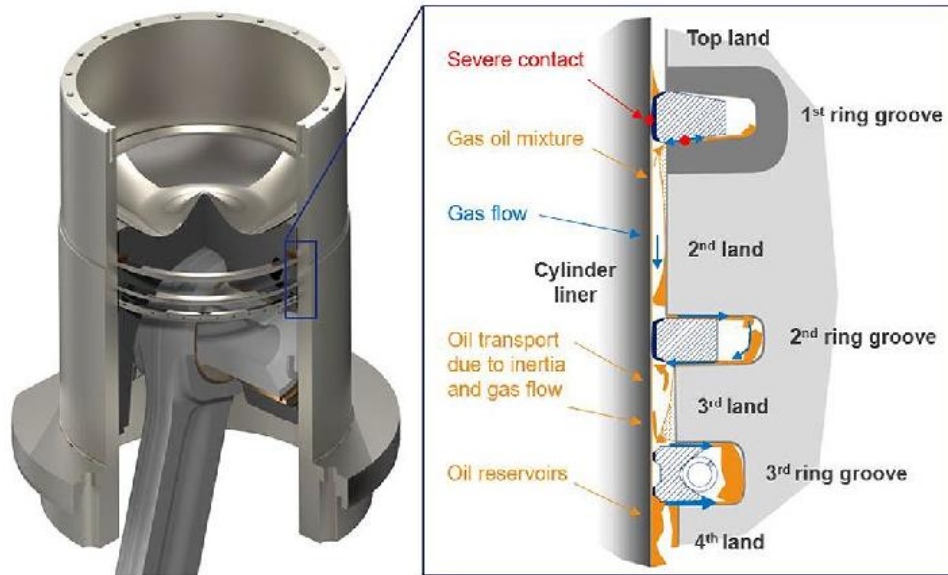
Lubrication of the cylinder liner and piston rings is primarily done by splash oil from the revolving motion of the crankshaft and oil vapor from the crankcase.



Lubrication between piston and cylinder liner

Lubrication oil is present between the piston rings and the cylinder liner to maintain a sealing effect for the combustion chamber.

Otherwise, a "blow by" phenomenon occurs, in which combustion gases leak into the crank chamber.



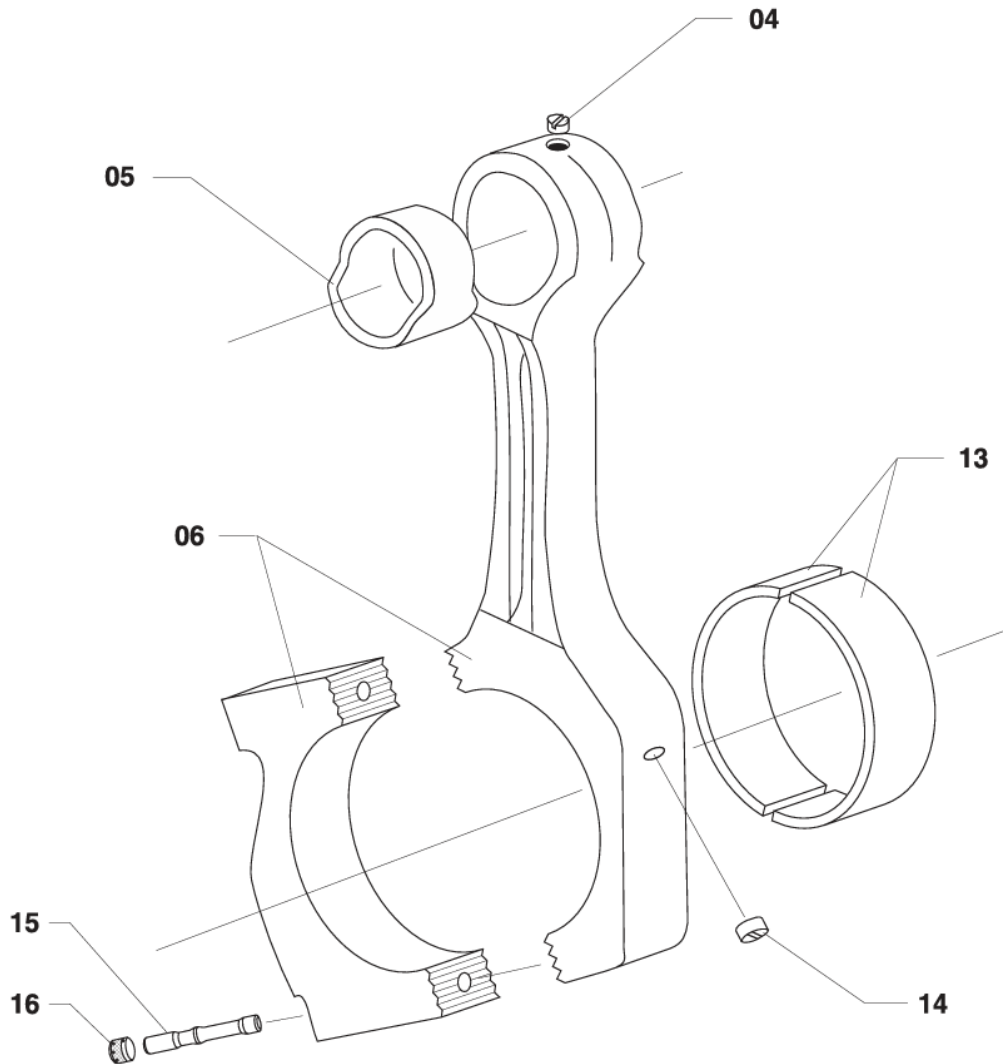
Piston



Connecting rod

The connecting rod is die-forged. The big-end has an inclined joint in order to facilitate the piston and connecting rod assembly to be withdrawn up through the cylinder liner. The joint faces on the connecting rod and the bearing cap are serrated to ensure precise location and to prevent relative movement of the parts.

04	1/C	Plug screw	13	1/C	Connecting rod bearing 2/2
05	1/C	Bush for connecting rod	14	1/C	Plug screw
06	1/C	Connecting rod incl. item Nos 04, 05, 14, 15, 16	15	2/C	Connecting rod stud
			16	2/C	Nut



Connecting rod



2 connecting rod bolts



Connecting rod



4 connecting rod bolts

Connecting rod

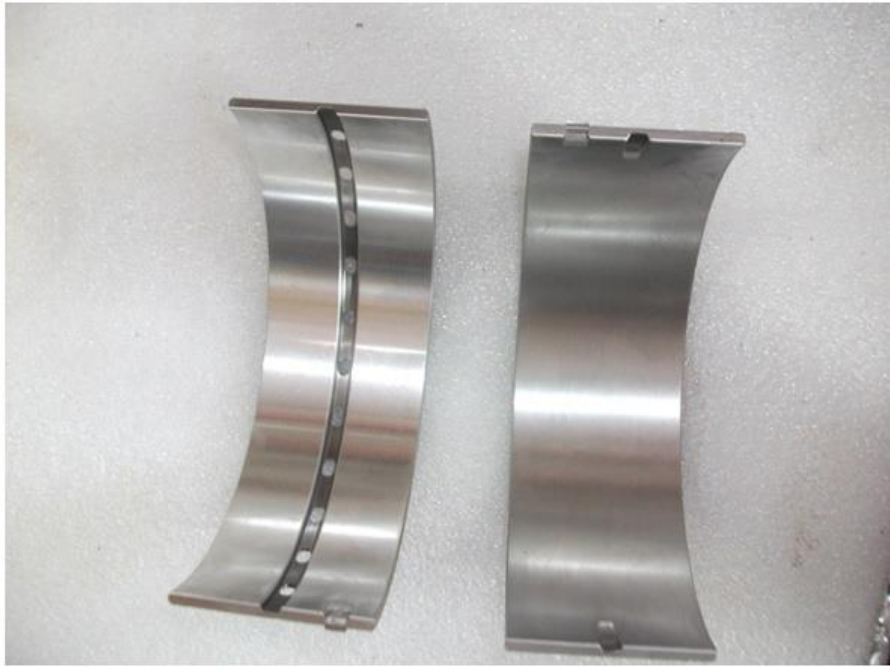


Some G/E models use 4 studs and 8 hydraulic nuts for connecting rod



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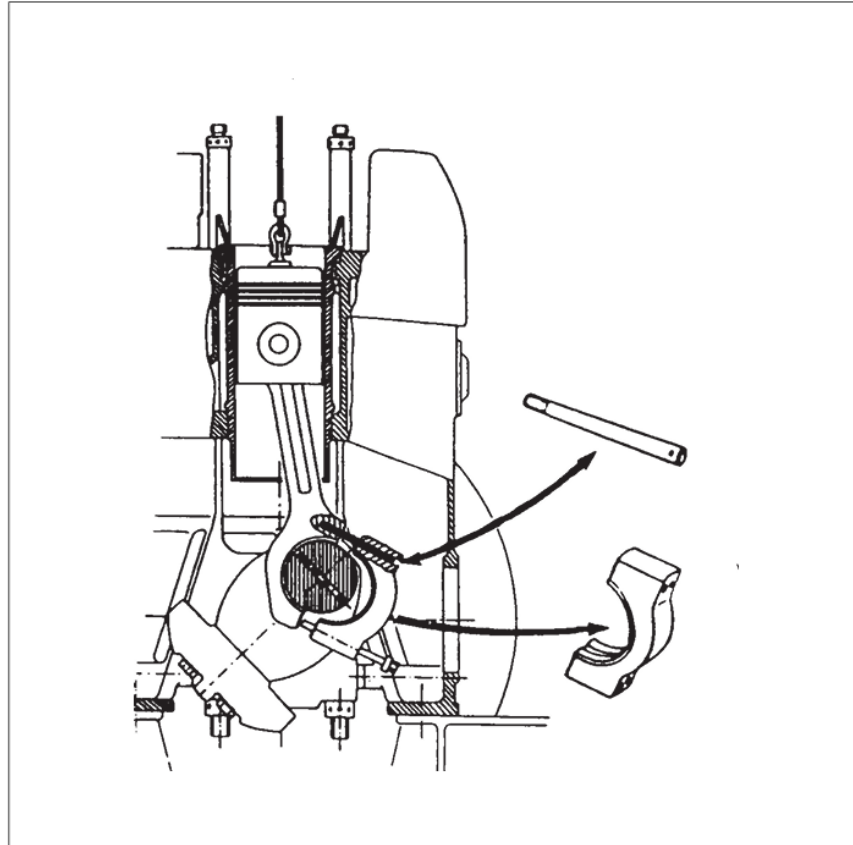
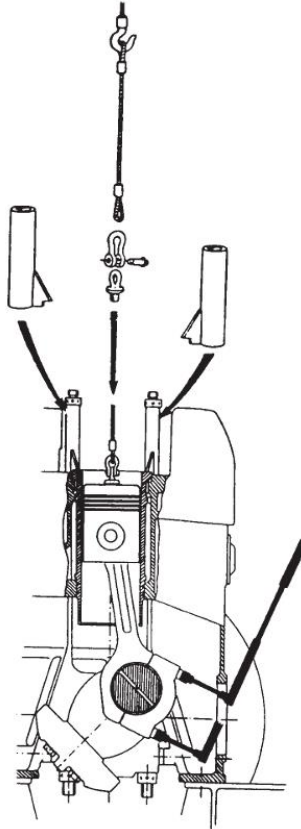
Connecting rod bearing



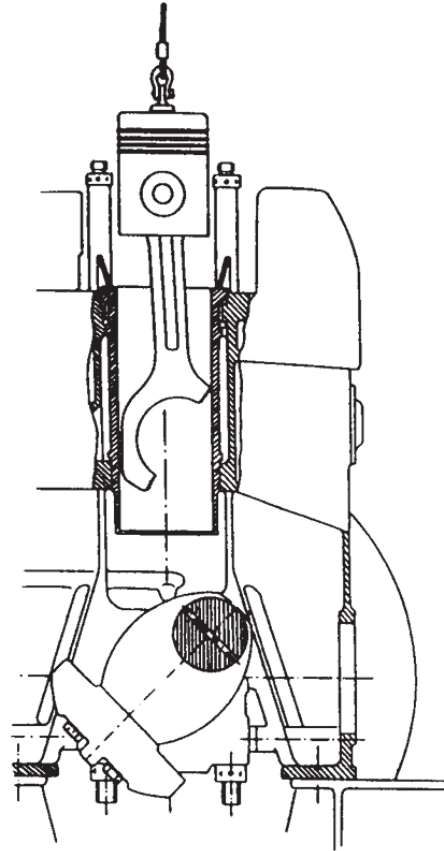
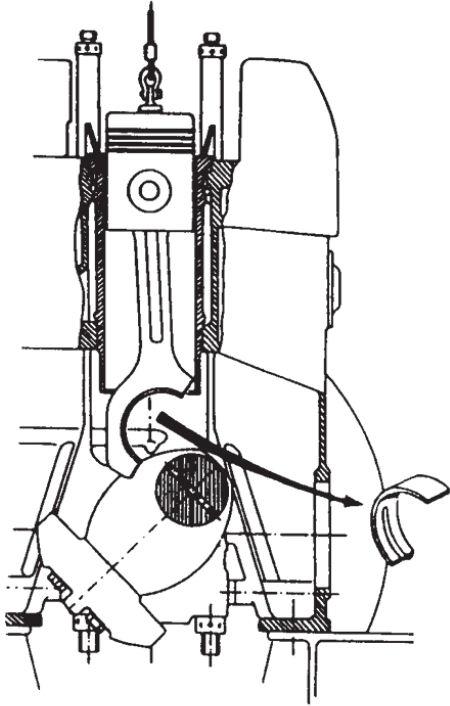
Connecting rod bearing



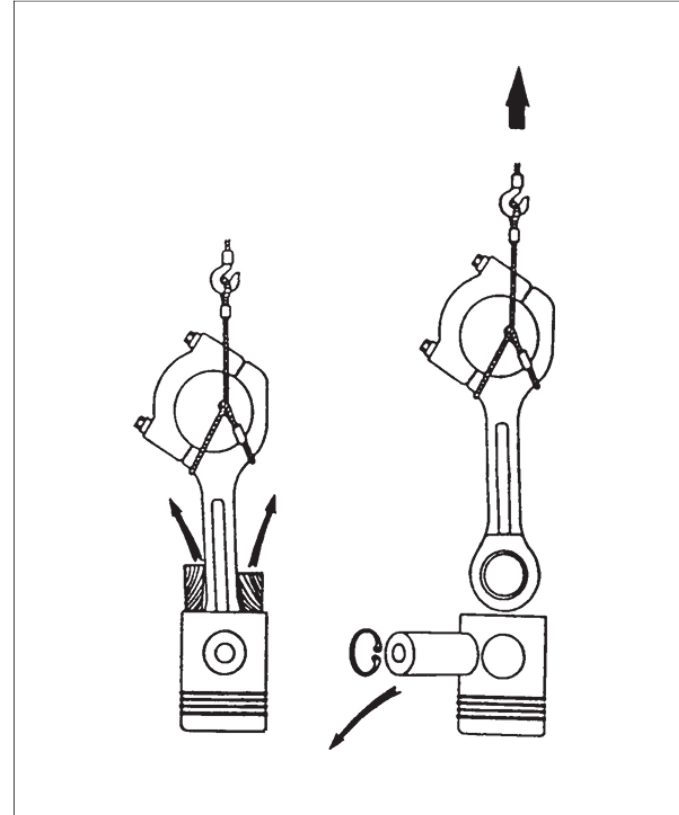
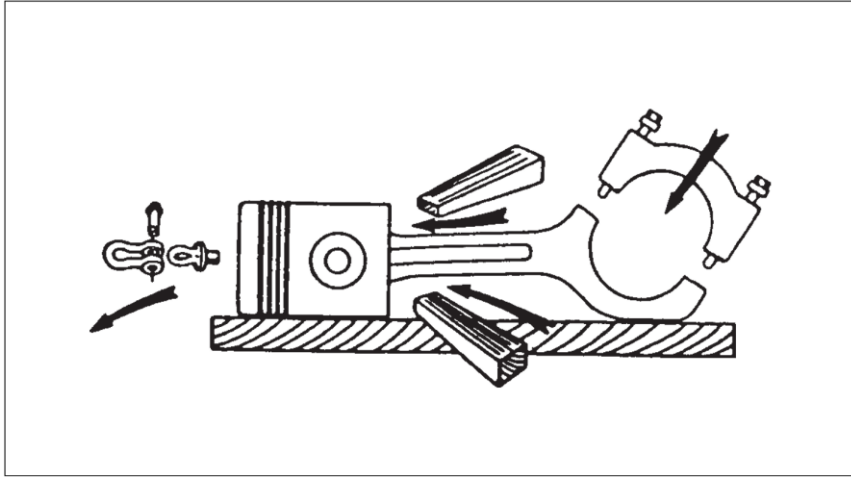
Dismounting of piston and connecting rod



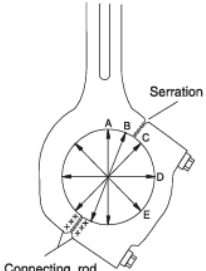
Dismounting of piston and connecting rod



Separation of piston and connecting rod

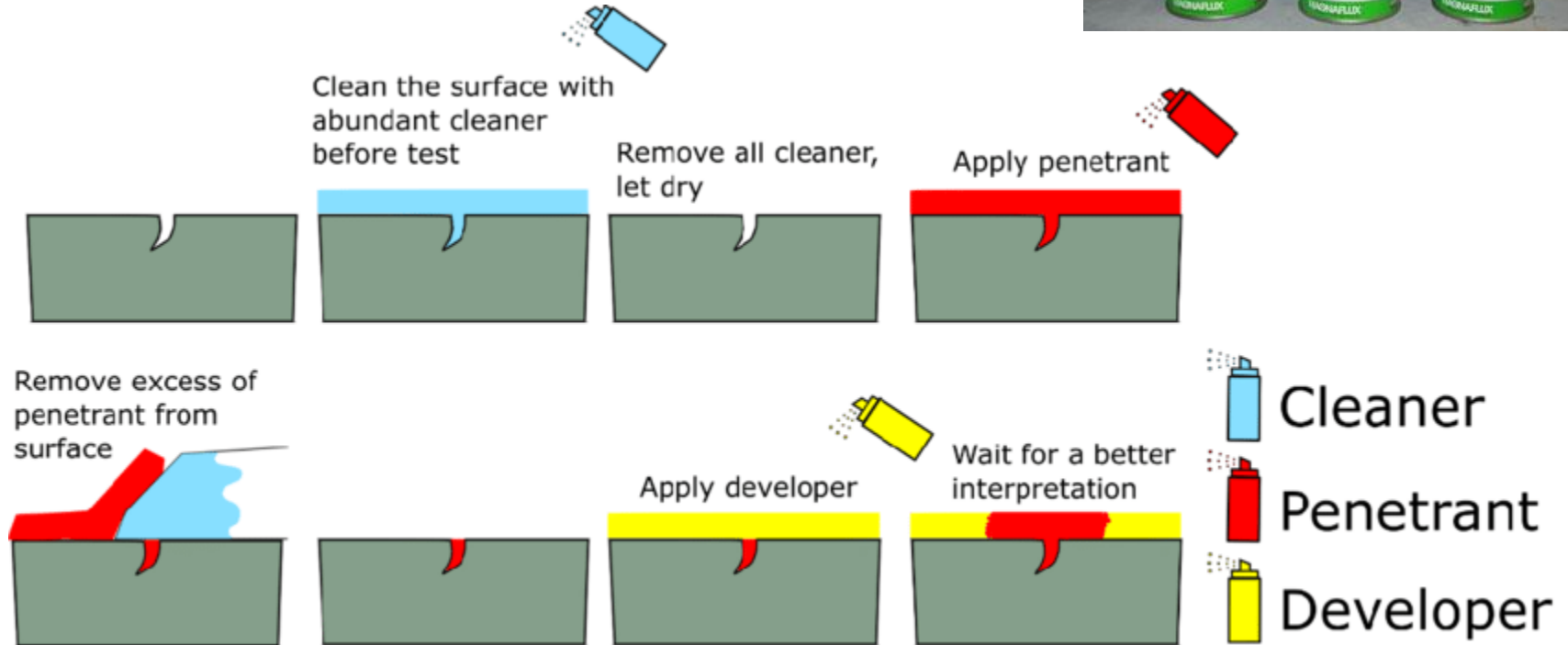


Connecting rod inspection

Connecting Rod Inspection for L23/30H				Plant/Ship:		Engine Type:		
				Engine No.:		Running Hours:		
				Sign.:		Insp. Date:		
Cylinder No. Connecting rod ident. No. Running hours for connecting rod	1	2	3	4	5	6	7	8
								
Ovalness: Diff. between min./max. Condition of serration								
Tightening for measurement see instruction. 	☐ Serration OK ☐ Wear ☐ Cracks ☐ Corrosion/ Pitting ☐ Impact mark ☐ ☐ ☐ Remarks:	☐ Serration OK ☐ Wear ☐ Cracks ☐ Corrosion/ Pitting ☐ Impact mark ☐ ☐ ☐ Remarks:	☐ Serration OK ☐ Wear ☐ Cracks ☐ Corrosion/ Pitting ☐ Impact mark ☐ ☐ ☐ Remarks:	☐ Serration OK ☐ Wear ☐ Cracks ☐ Corrosion/ Pitting ☐ Impact mark ☐ ☐ ☐ Remarks:	☐ Serration OK ☐ Wear ☐ Cracks ☐ Corrosion/ Pitting ☐ Impact mark ☐ ☐ ☐ Remarks:	☐ Serration OK ☐ Wear ☐ Cracks ☐ Corrosion/ Pitting ☐ Impact mark ☐ ☐ ☐ Remarks:	☐ Serration OK ☐ Wear ☐ Cracks ☐ Corrosion/ Pitting ☐ Impact mark ☐ ☐ ☐ Remarks:	☐ Serration OK ☐ Wear ☐ Cracks ☐ Corrosion/ Pitting ☐ Impact mark ☐ ☐ ☐ Remarks:

Note! The ident. No. on the connecting rod and on the bearing cap must always be the same.

Liquid penetrant test



Liquid penetrant test



Criteria for replacement of connecting rod big-end and main bearing shells (only reference)

This paper gives information about the evaluation of the connecting rod big-end and main bearing shells when wear appears on the running surface under normal operating conditions.

Bearing damages caused by incorrect running conditions, like

- Scoring of the running surface due to foreign particles in the lubricating oil.
- Cavitation.
- Corrosion
- Overloading, overheating a.s.o.

are not described in this paper.

In these cases, the bearing shells must be exchanged, of course, and in order to avoid further bearing failures, the cause of the failure must be found and eliminated.

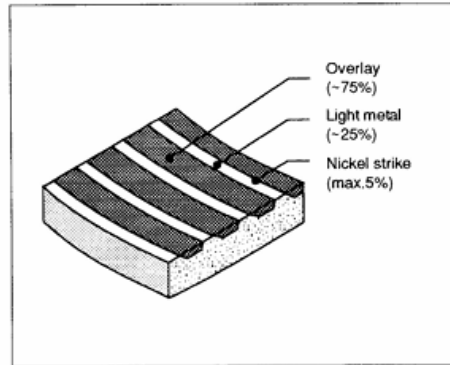


Fig 1. Running surface of the bearing shell.

A new bearing shell has a running surface with approx. 75 % galvanic overlay and approx. 25 % light-alloy metal, see fig 1.

Wear of the bearings running surface starts primarily in the soft overlay, where the overlay in the slots is reduced approx. 0.001 mm. The difference between the light-alloy metal and the overlay is by further wear almost constant, i.e. approx. 0.005 mm.

The gravity of the running surface's wear must be determined with a magnifier (x30), hereby the overlay can be seen as the dark area and the light-alloy metal as the light area.

The ratio between the width of the light-alloy metal and the width of the slot as well as the dimension of the worn area are important for determining the wear of the bearing.

The bearing can also operate although part of the overlay is missing in the slots. Practice has shown, that bearings with partly empty slots can run without any negative influence on the operation of the bearing.

By determining the condition of the bearing shell, the overlay at a minor stressed area must be used as comparison.

The valuation of the bearing shell condition is divided into 5 groups:

1. Re-use
2. Limiting case - wear
3. Limiting case - local levelling
4. Limiting case - permanent breakage on the overlay
5. Empty slots.

In case of a combination of the above mentioned conditions, the lowest possible limits are valid.



1.1. Bearing Shells for Re-use.

Condition: The slot geometry corresponds to a new bearing shell. The overlay in the slots is fully preserved, see fig 2.

Dark spots are mainly oil coke remains.

Valuation: The ratio between light-alloy metal and overlay is 75 % to 25 %. Re-use!

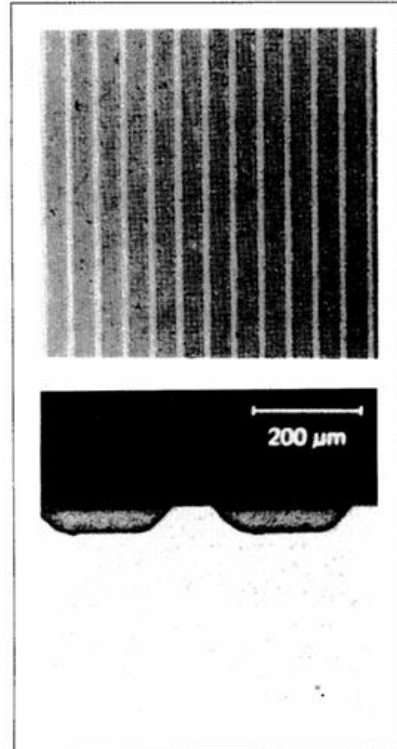


Fig 2.

1.2. Bearing Shells for Re-use.

Condition: The overlay is equally worn, approx. 0.005 mm in the slots. The light-alloy strips show no wear, see fig 3.

Dark spots are mainly oil coke remains.

Valuation: Due to worn running surface the light-alloy strips are slightly widened. Re-use!

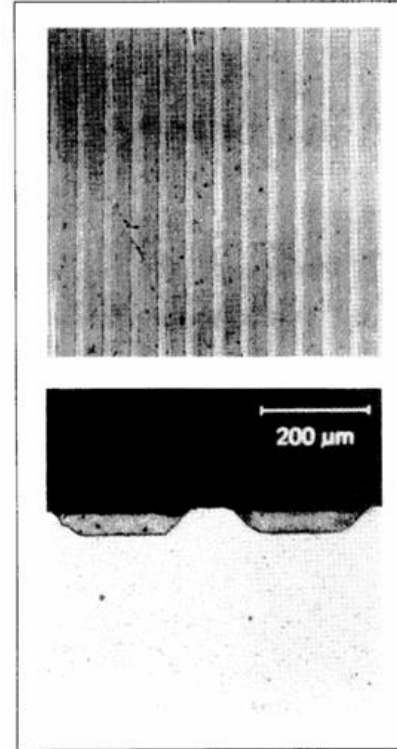
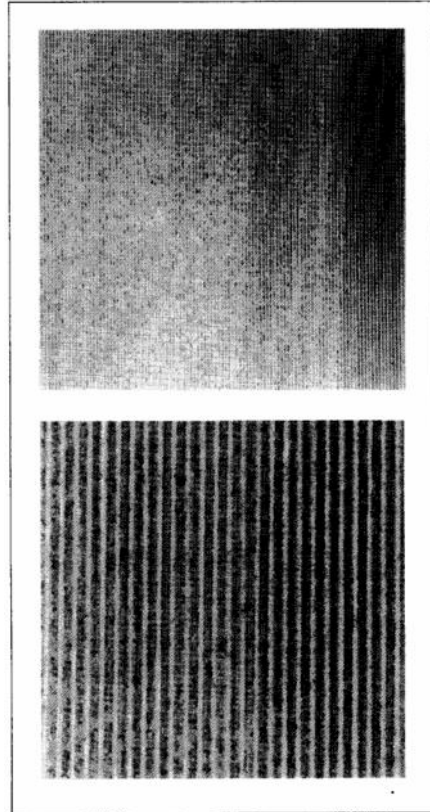


Fig 3.

1.3. Bearing Shells for Re-use.

Condition: Small particles all over the overlay. The light-alloy metal shows no changes of importance, see fig 4.

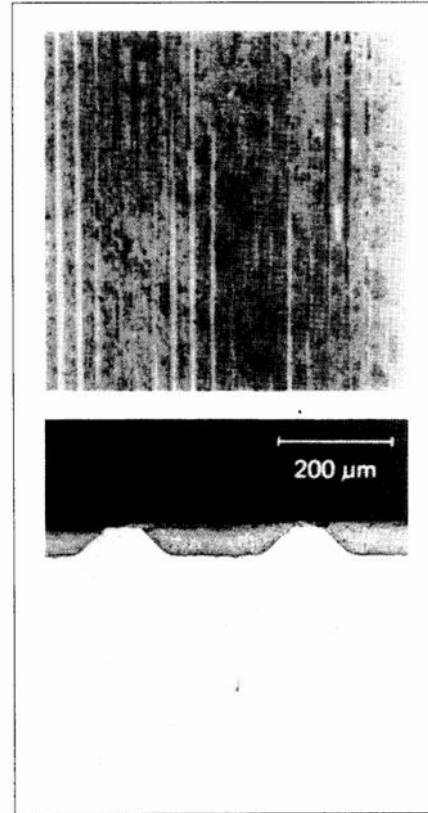
Valuation: Re-use of the shell, as the particles are wedged in the slots.



1.4. Bearing Shells for Re-use.

Condition: The overlay has been dragged and smudged over the light-alloy metal. These are only partly visible, see fig 5.

Valuation: Re-use of the shell.



2. Limiting Case - Wear

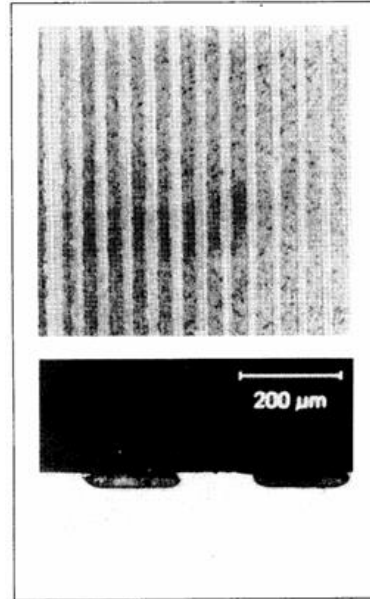


Fig 6.

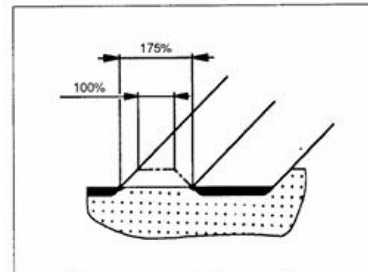


Fig 7.

Condition: The bearing shell is locally so worn, that the ratio between the light-alloy metal and the overlay is 1:1, see fig 6. The width of the light-alloy metal has increased from 100 % (new bearing shell) up to 175 %. In the slots some overlay is still left, see fig 7.

Valuation: The bearing can operate with wear of the below listed size, see fig 8 og 9. If the wear limits are exceeded the bearing shells must be exchanged due to safety precautions.

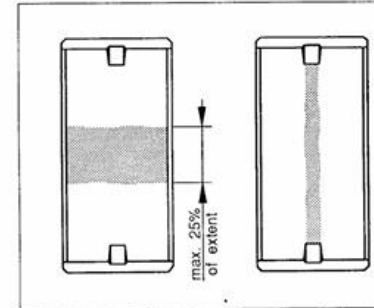


Fig 8.

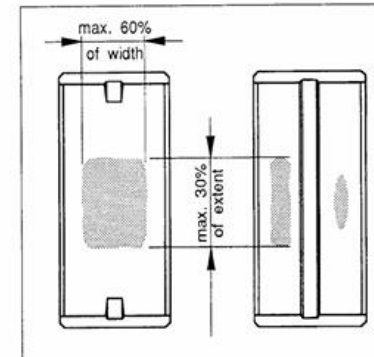


Fig 9.

3. Limit Case - Levelling

Condition: The slots are locally totally levelled as shown in fig 10.

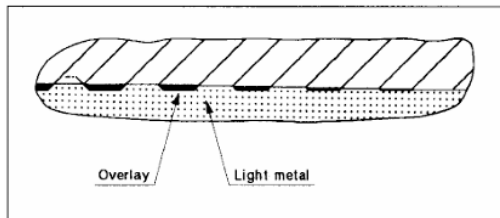


Fig 10.

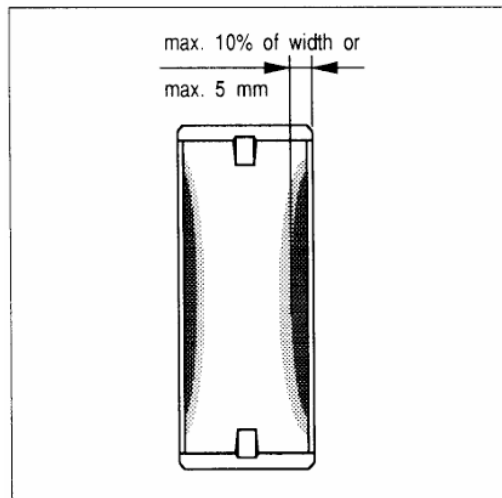


Fig 11.

Valuation: If the limits shown in fig 11 and 12 are exceeded the bearing shells must be exchanged.

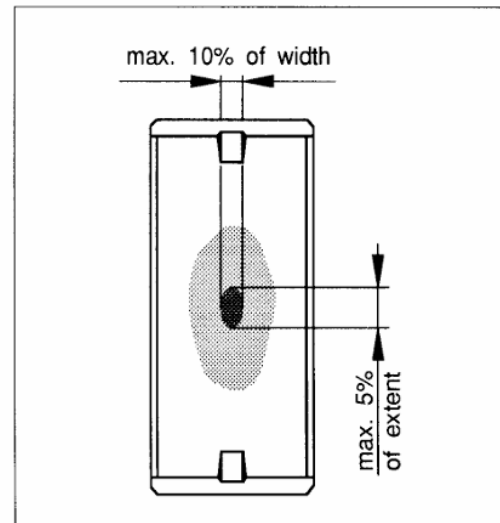


Fig 12.

4. Limit Case - Permanent Breakage of the Running Surface

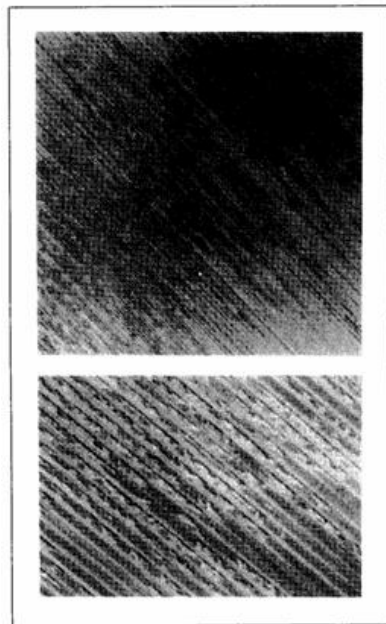


Fig 13.

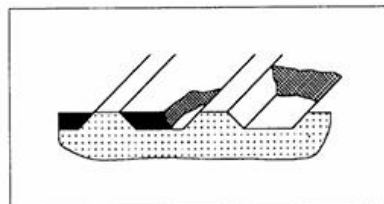


Fig 14.

Condition: Breakage in the overlay due to local overload, as shown in fig 13 and 14.

Valuation: If the limits as shown below in fig 15 and 16 below are exceeded with totally or partially empty slots, the bearing shells must be exchanged.

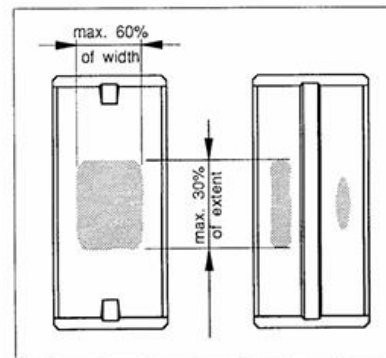


Fig 15.

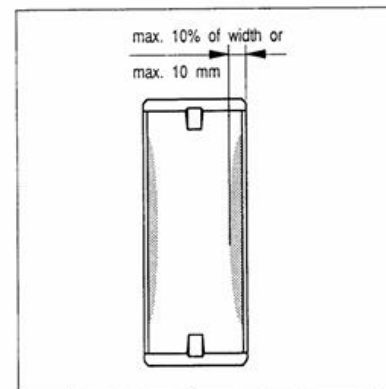


Fig 16.

5. Limit Case - Empty Slots

Condition: In some areas there is no overlay in the slots, see fig 17.

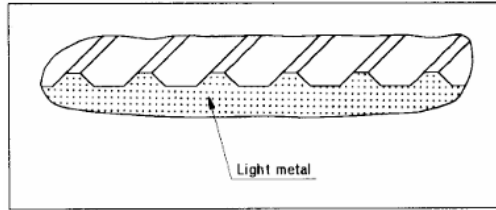


Fig 17.

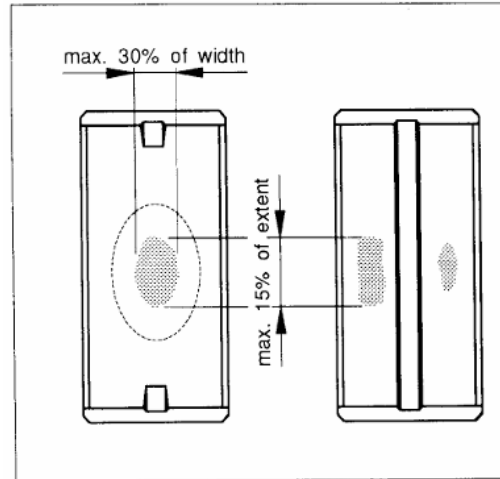


Fig 18.

Valuation: If the limits shown in the below fig 18 and 19, are exceeded, the bearing shells must be exchanged.

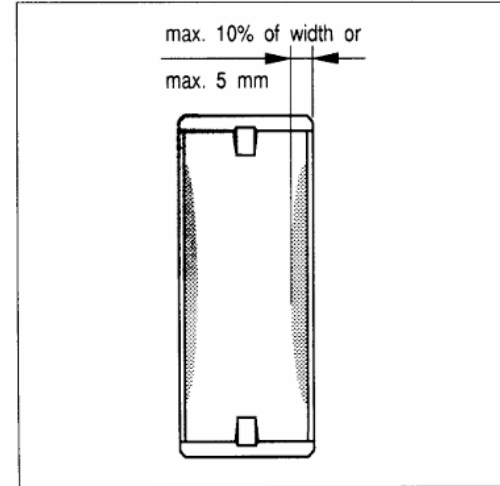
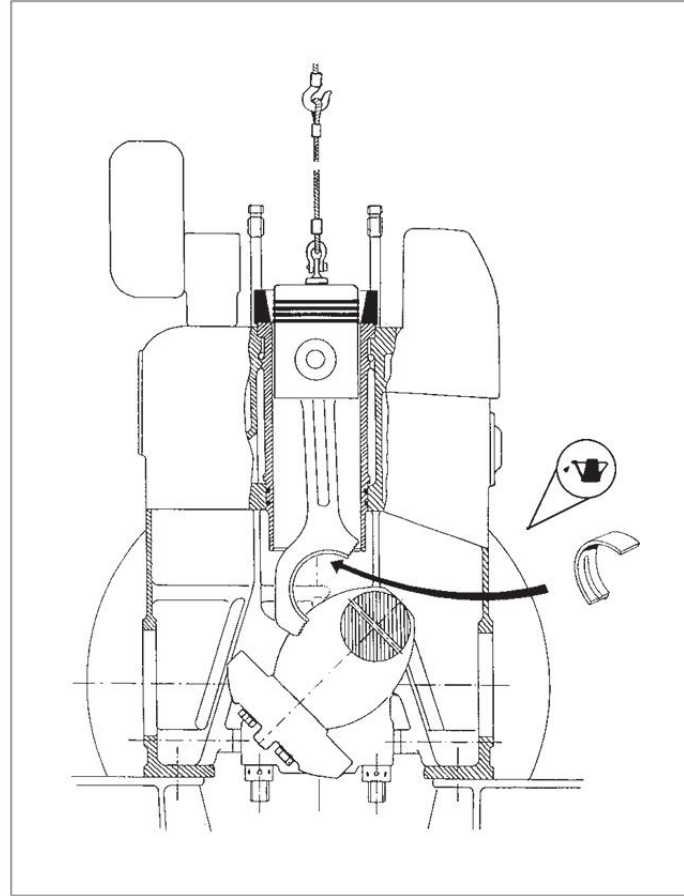
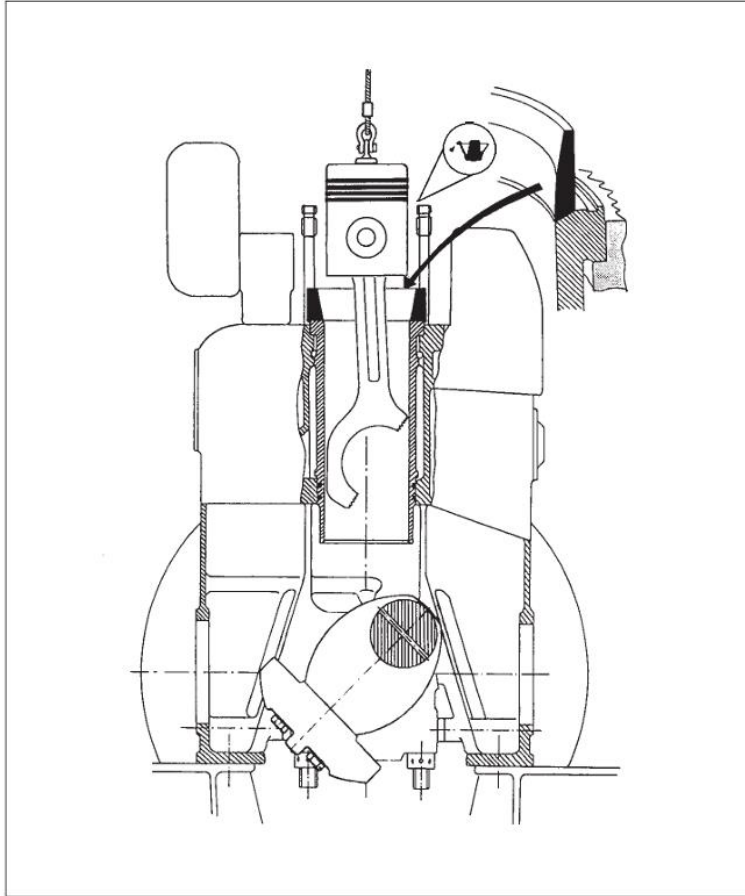
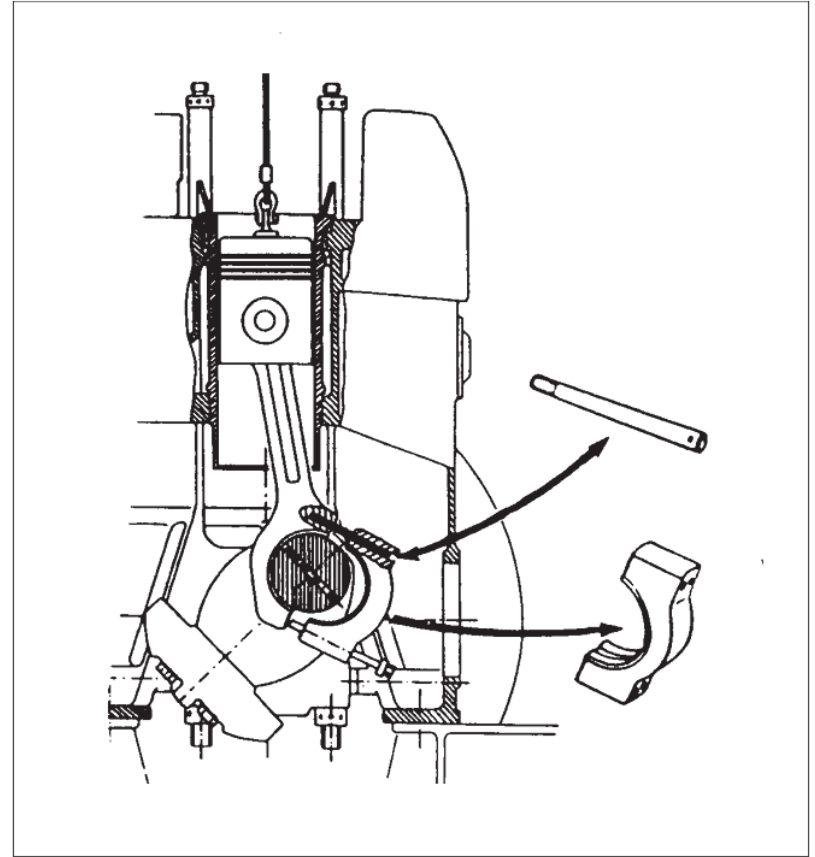
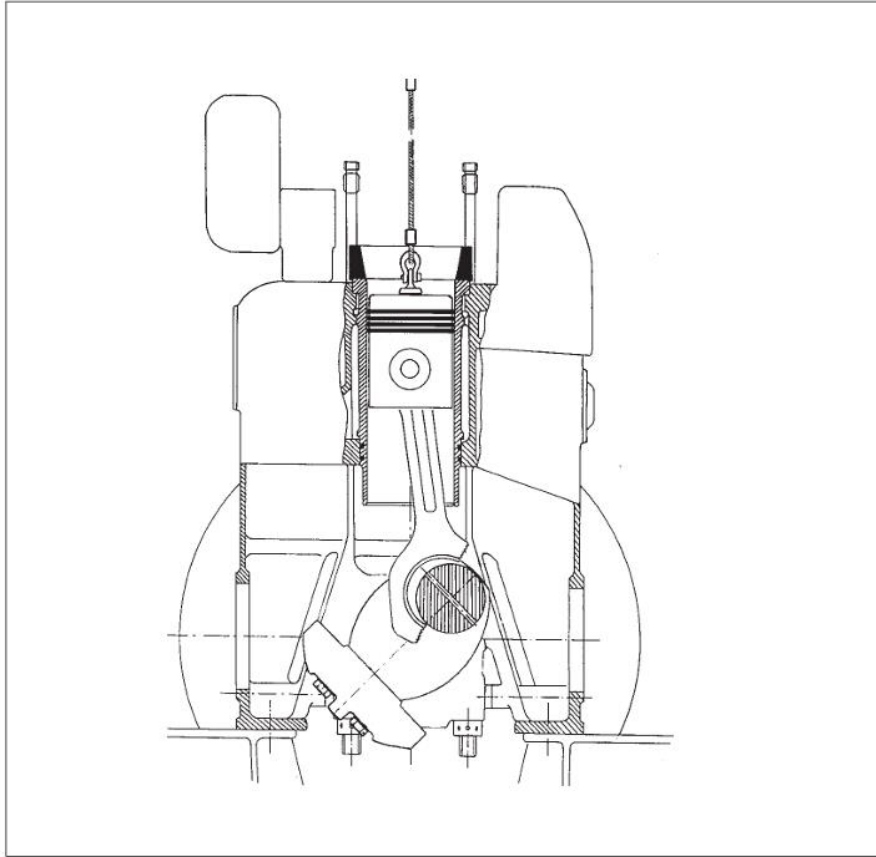


Fig 19.

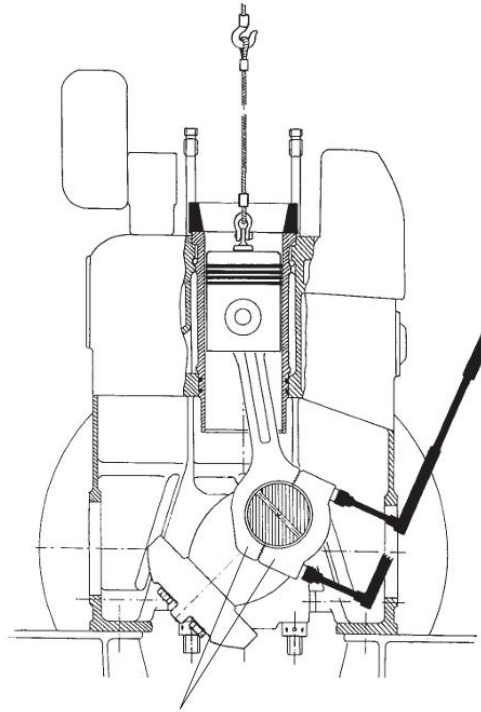
Mounting of piston and connecting rod



Mounting of piston and connecting rod



Mounting of piston and connecting rod



Con. Rod - Ident No.



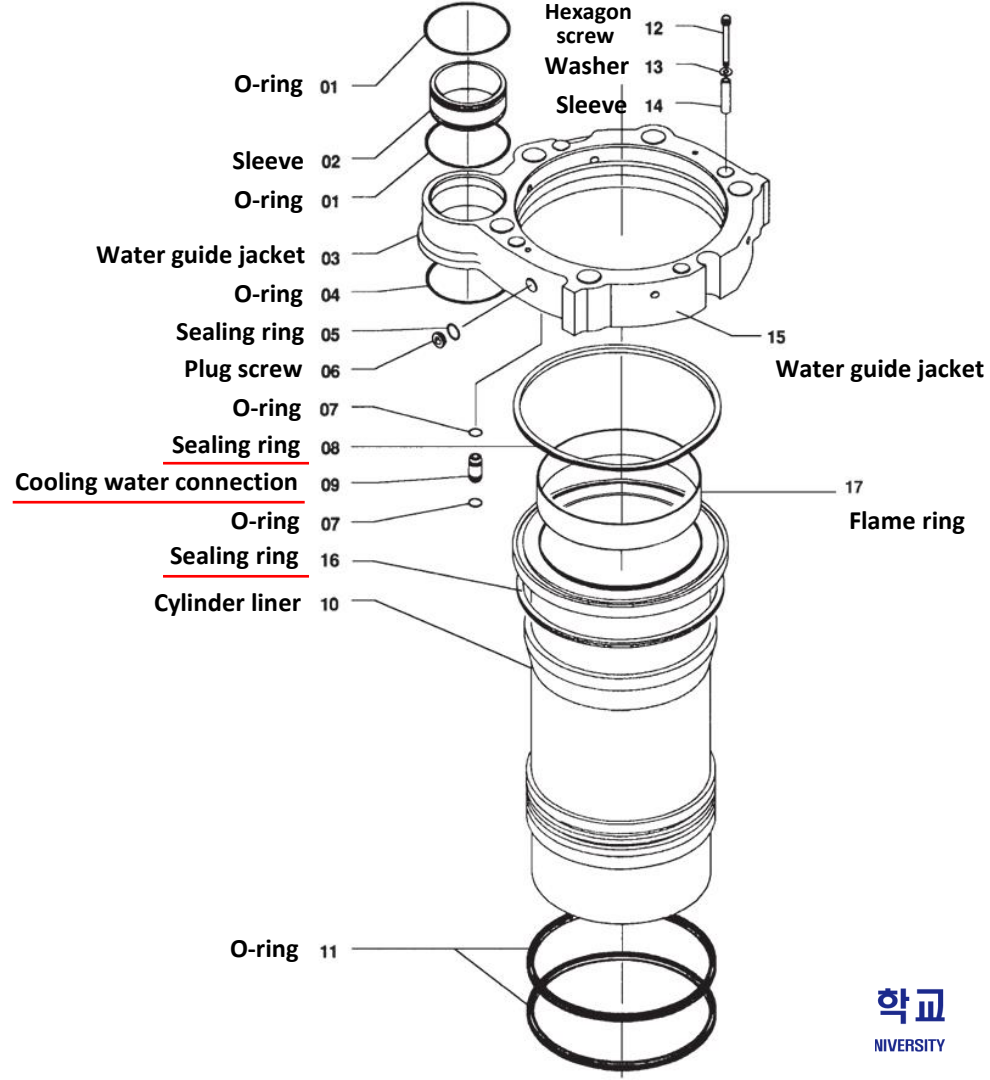
Cylinder liner

The cylinder liner is made of fine-grained, pearlite cast iron and is fitted in a bore in the engine frame.

Replaceable cast iron sealing rings are fitted between the liner and the cylinder head and between the liner and the frame.

The liner is clamped by the cylinder head and is guided by a bore at the bottom of the cooling water space of the engine frame. The liner can thus expand freely downwards, when heated during the running of the engine.

Sealing for the cooling water is obtained by means of silicone rubber rings which are fitted in grooves machined in the liner.



Cylinder liner

From the top view



2 o-rings are inserted here



1 sealing ring is placed here



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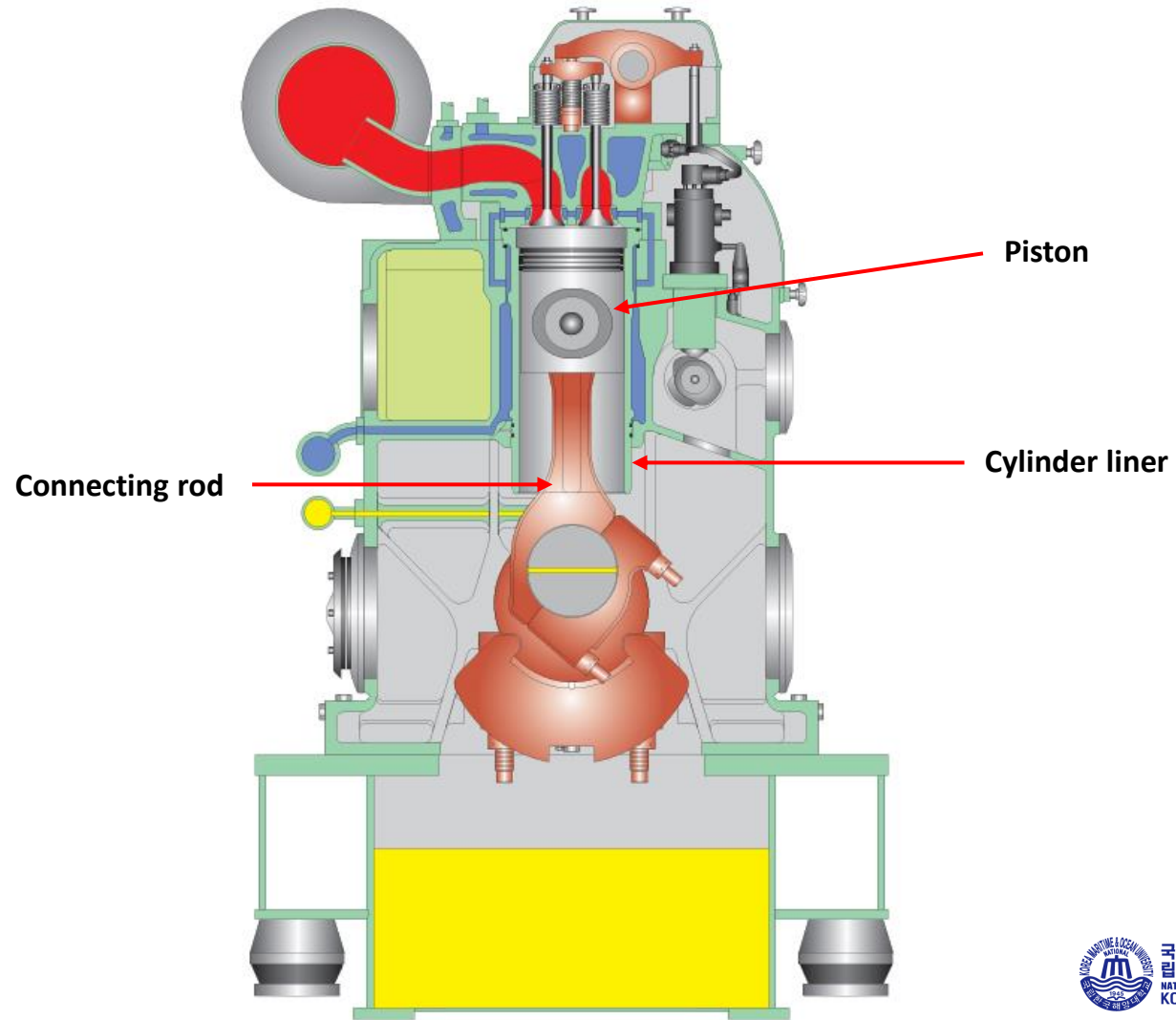
Cylinder liner



Chain block

If the inner wear of the cylinder liner is severe, it is removed and replaced with a new one, as shown in the photo on the left.

After fixing the chain block to the ceiling, hang the cylinder liner on the hook and slowly pull it up.



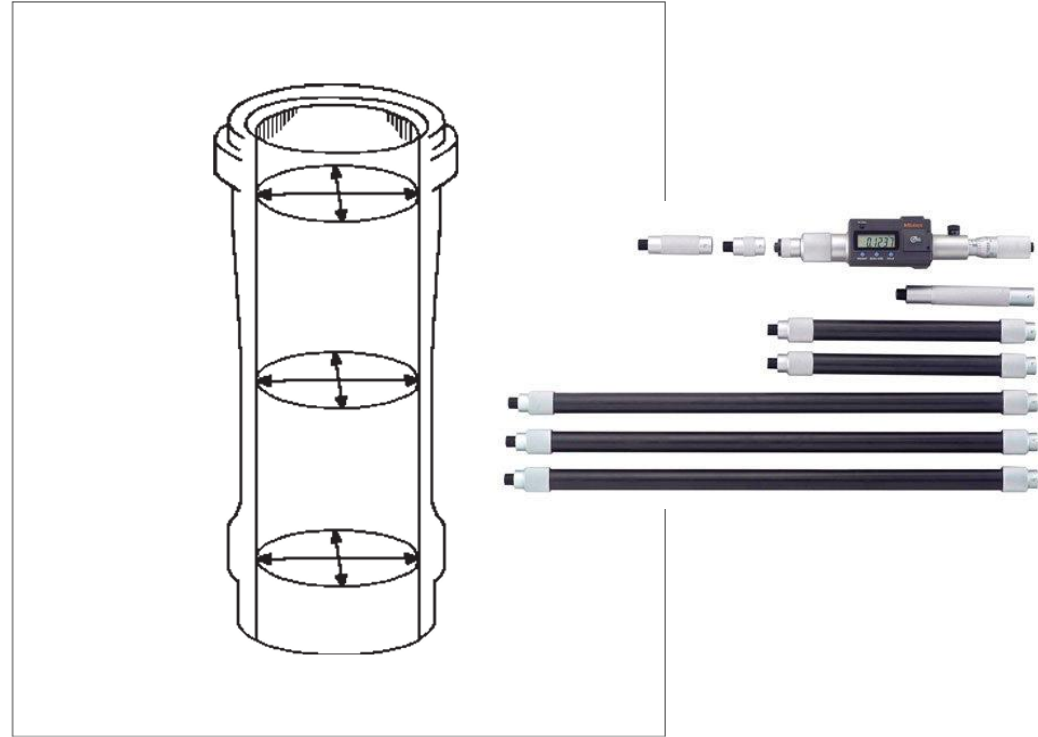
Inspection of cylinder liner

Measurement of Cylinder Diameter

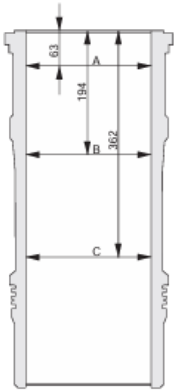
While the piston is removed from the cylinder, the latter is measured to record the wear.

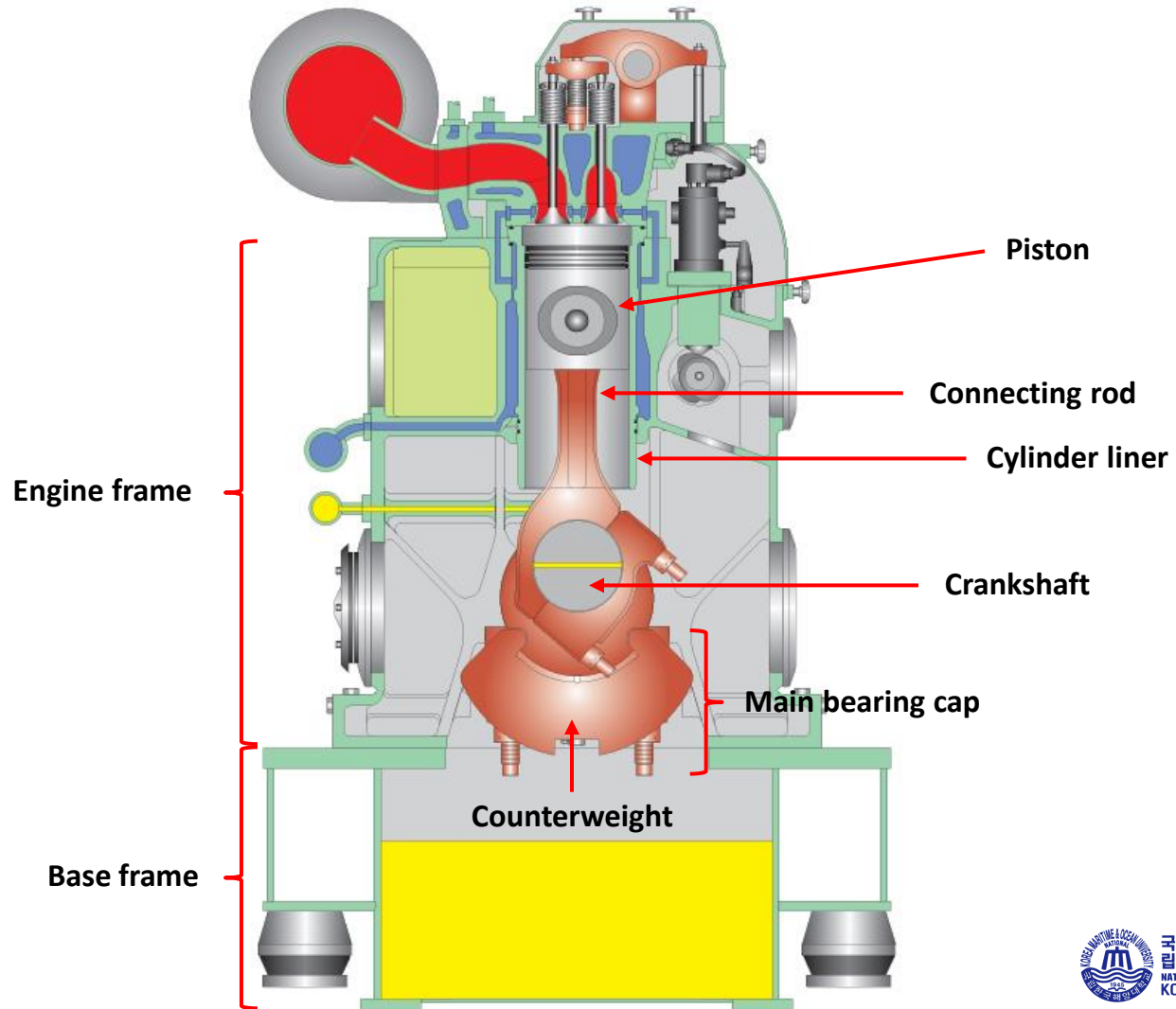
The measurements are taken by means of an inside micrometer, with measuring points at TDC-position for uppermost piston ring, halfway down and at the bottom of the cylinder liner.

The measurements should normally be taken in transverse as well as in longitudinal direction.

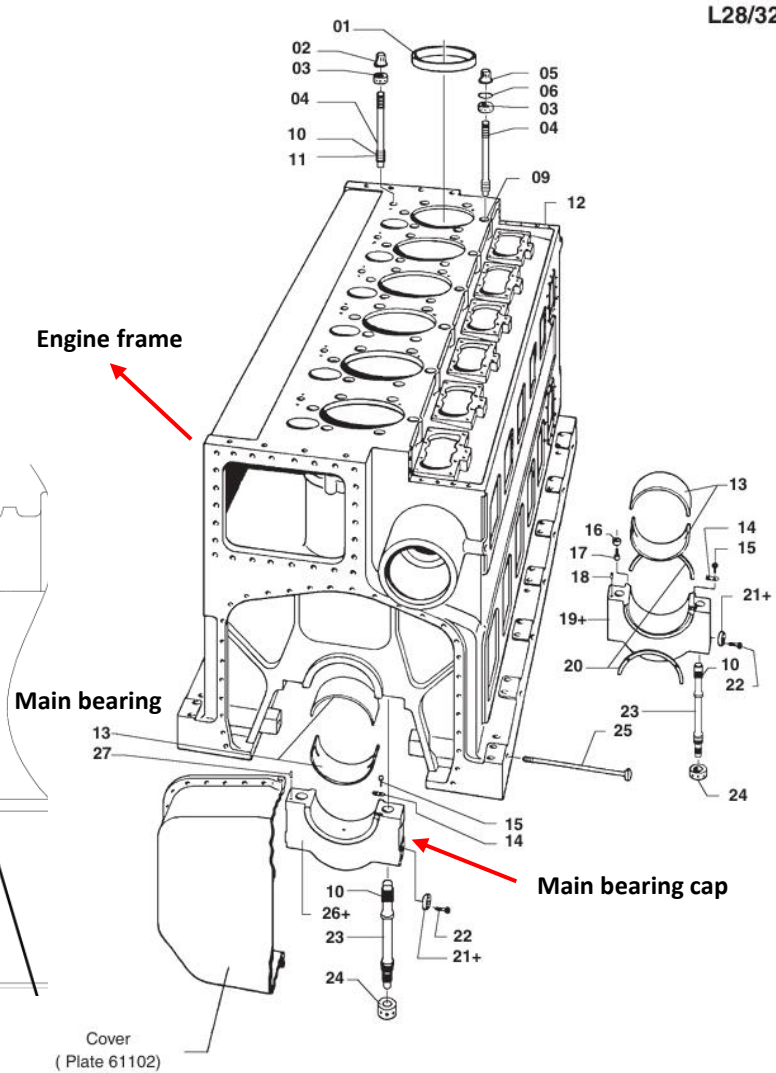
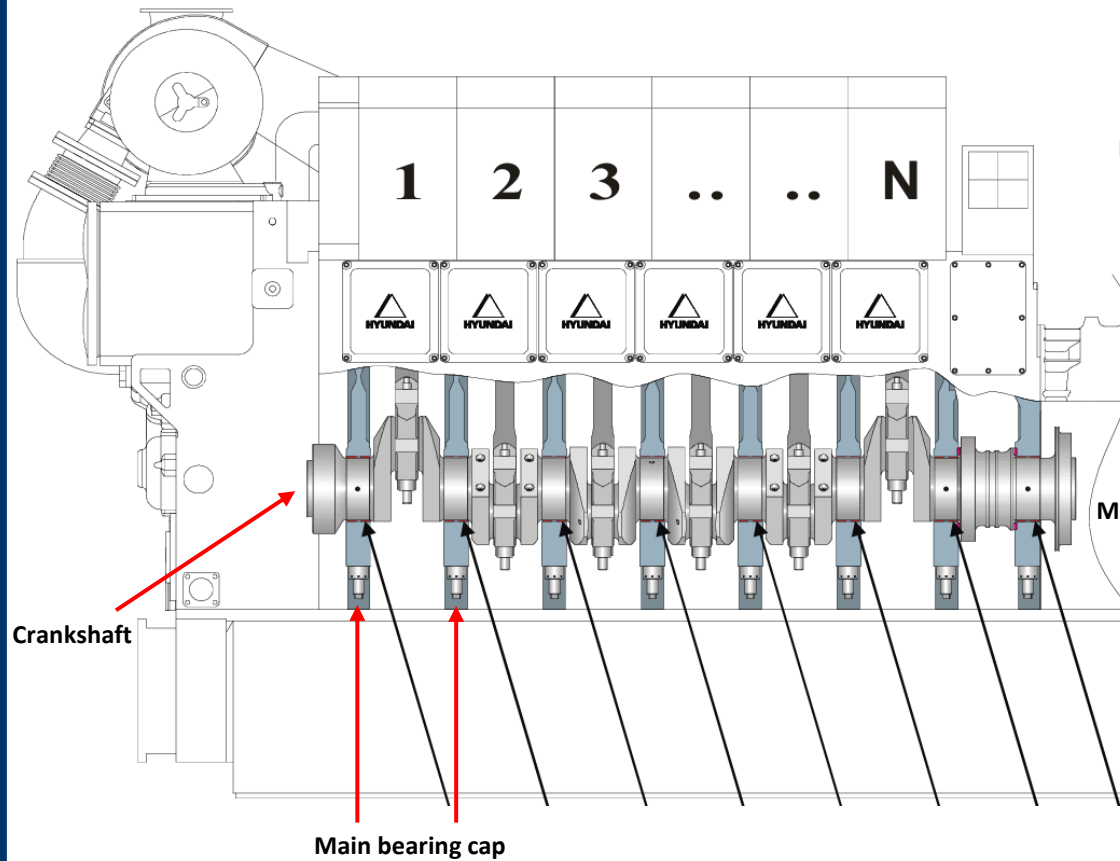


Inspection of cylinder liner

Measurement of Cylinder Liner				Engine type			Report No.		
Plant				Engine No.			Encl. No.		
				Running hours			Insp. date		
				Fuel			Sign.		
Cyl. No.	1	2	3	4	5	6	7	8	
Cylinder ident.No.									
Running hours									
	Lengthwise A B C Crosswise A B C								
		Liner temp. °C							
Remarks...									



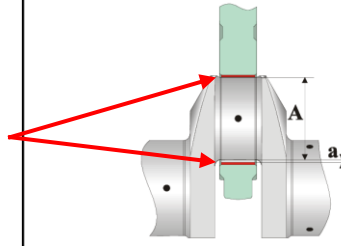
Crankshaft



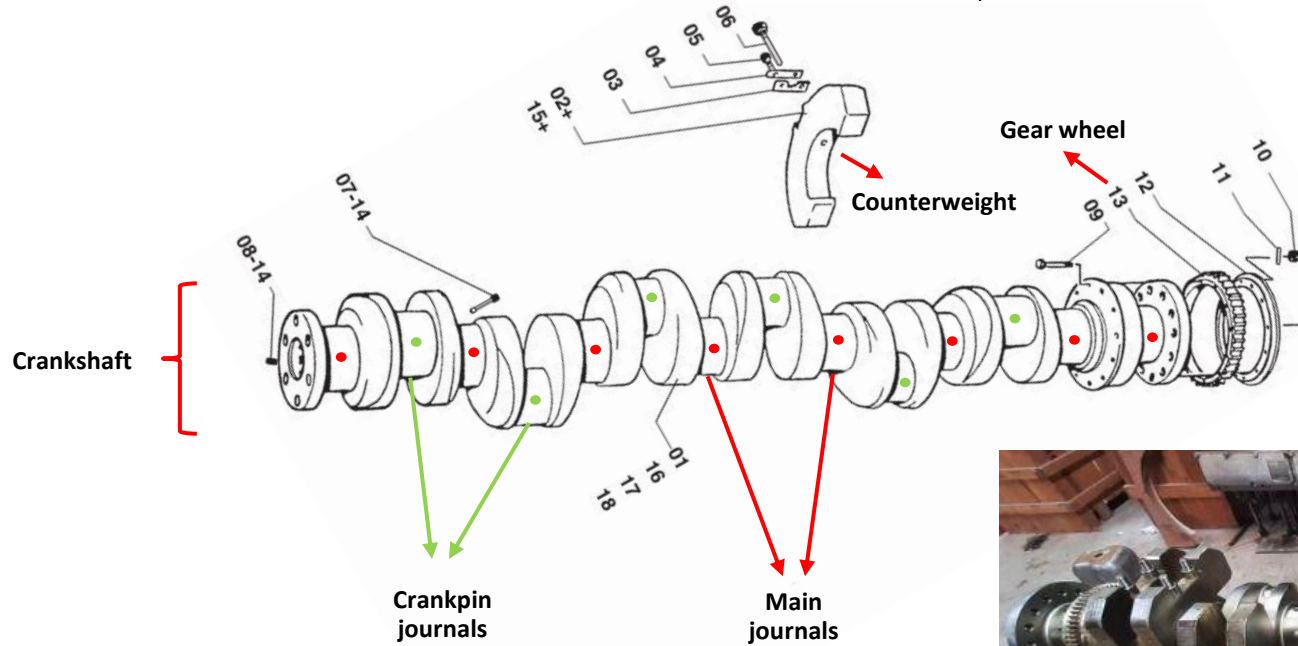
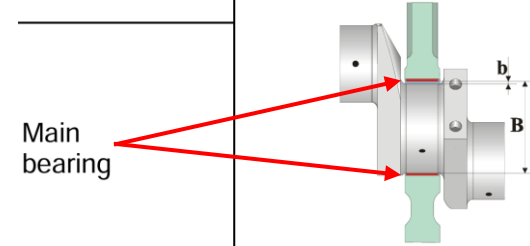
Crankshaft

Connecting rod bearing is also called as big-end bearing

Big-end bearing

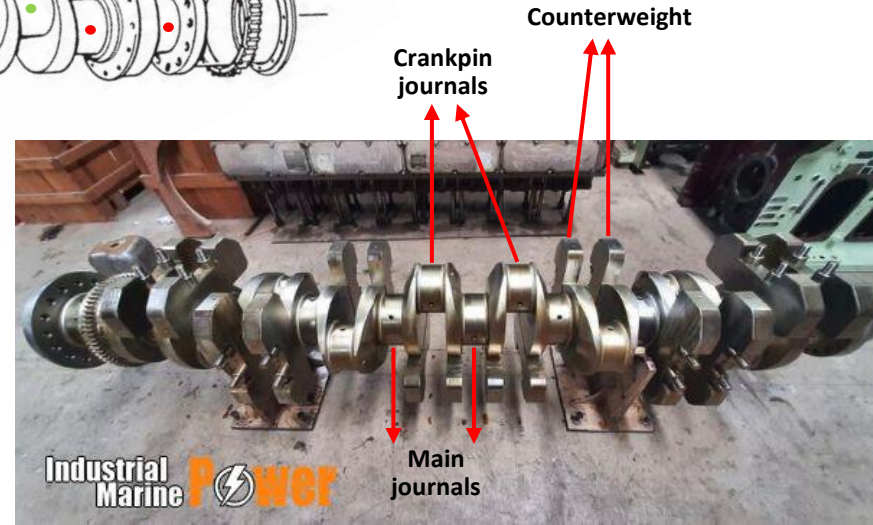


Main bearing



Two connecting rod bearings surround the crankpin journal, on which the connecting rod is installed.

Two main bearings surround the main journal, on which the engine frame and main bearing cap are installed.



Crankshaft



Crankshaft



Crankshaft

Counterweight

1. Offsetting Vibration and Unbalanced Forces

When the crankshaft, piston, and connecting rod move, not only rotational force but also vertical and horizontal unbalanced forces are generated. Counterweights counterbalance the centrifugal force generated by these unbalanced masses, thereby reducing engine vibration.

2. Reducing Crankshaft Load

The large inertial force generated by the piston's reciprocating motion directly impacts the crankshaft, resulting in excessive bearing load. The counterweight compensates for this, extending the life of the main bearings and crankpin.

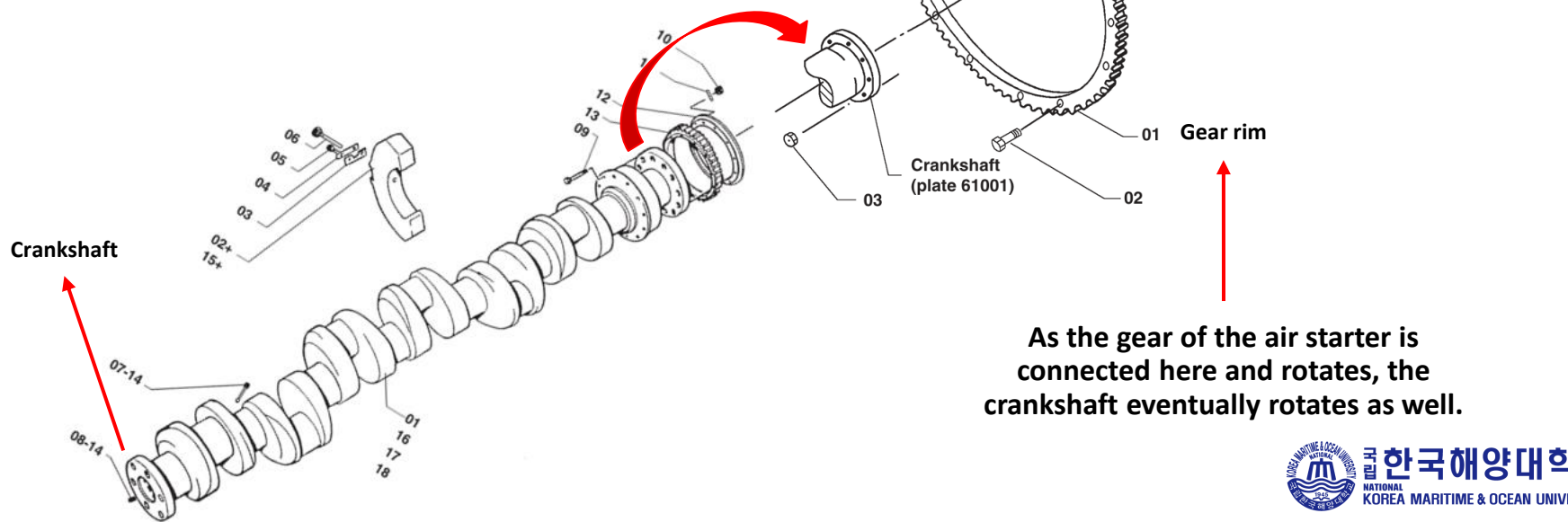
3. Maintaining Rotational Balance (Dynamic Balance)

Unbalanced masses at high speeds can cause severe vibration. Designing the counterweight with the appropriate location and weight maintains the engine's rotational balance.

4. Engine Stability and Noise Reduction

Balanced rotation reduces unnecessary vibration and noise.

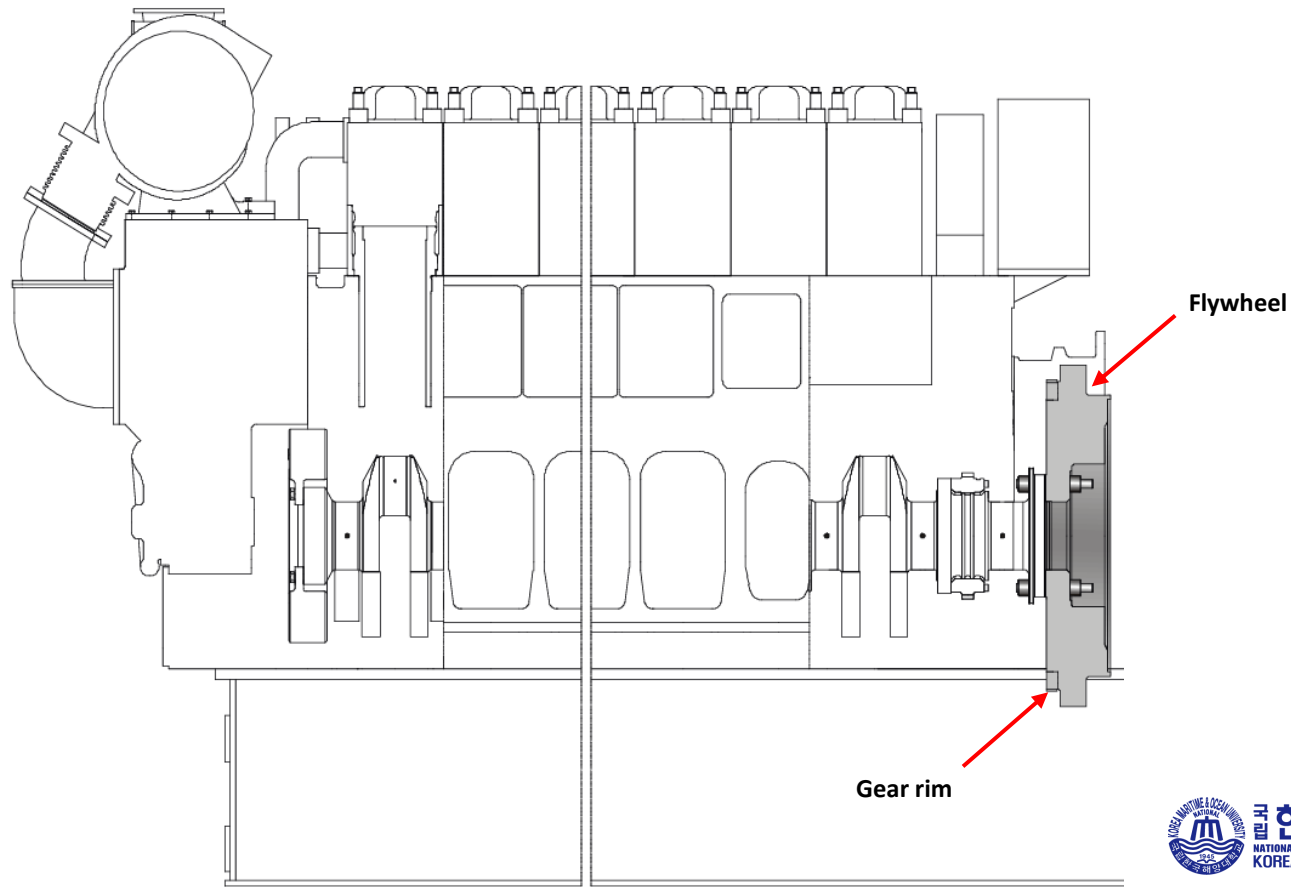
Crankshaft

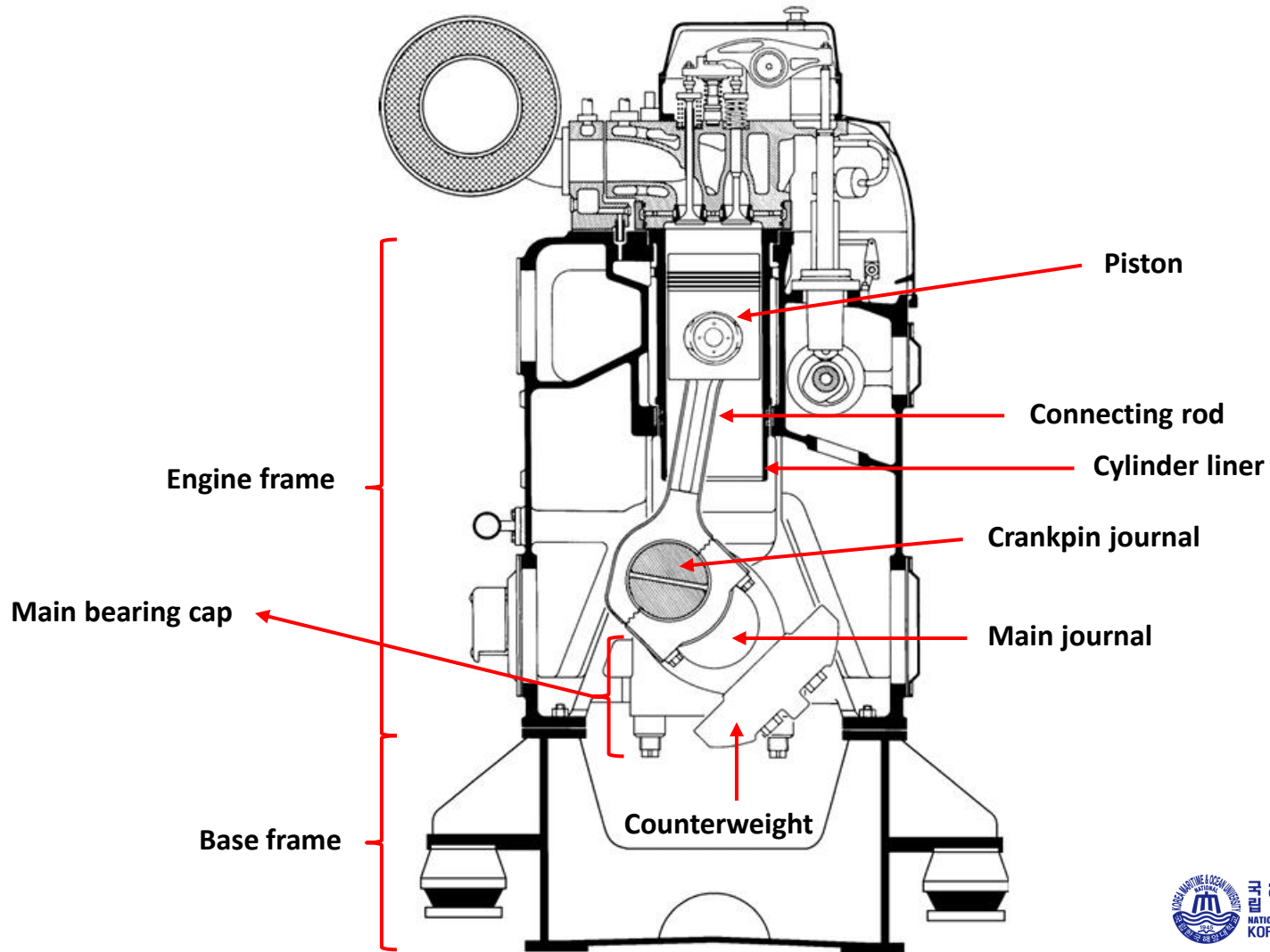


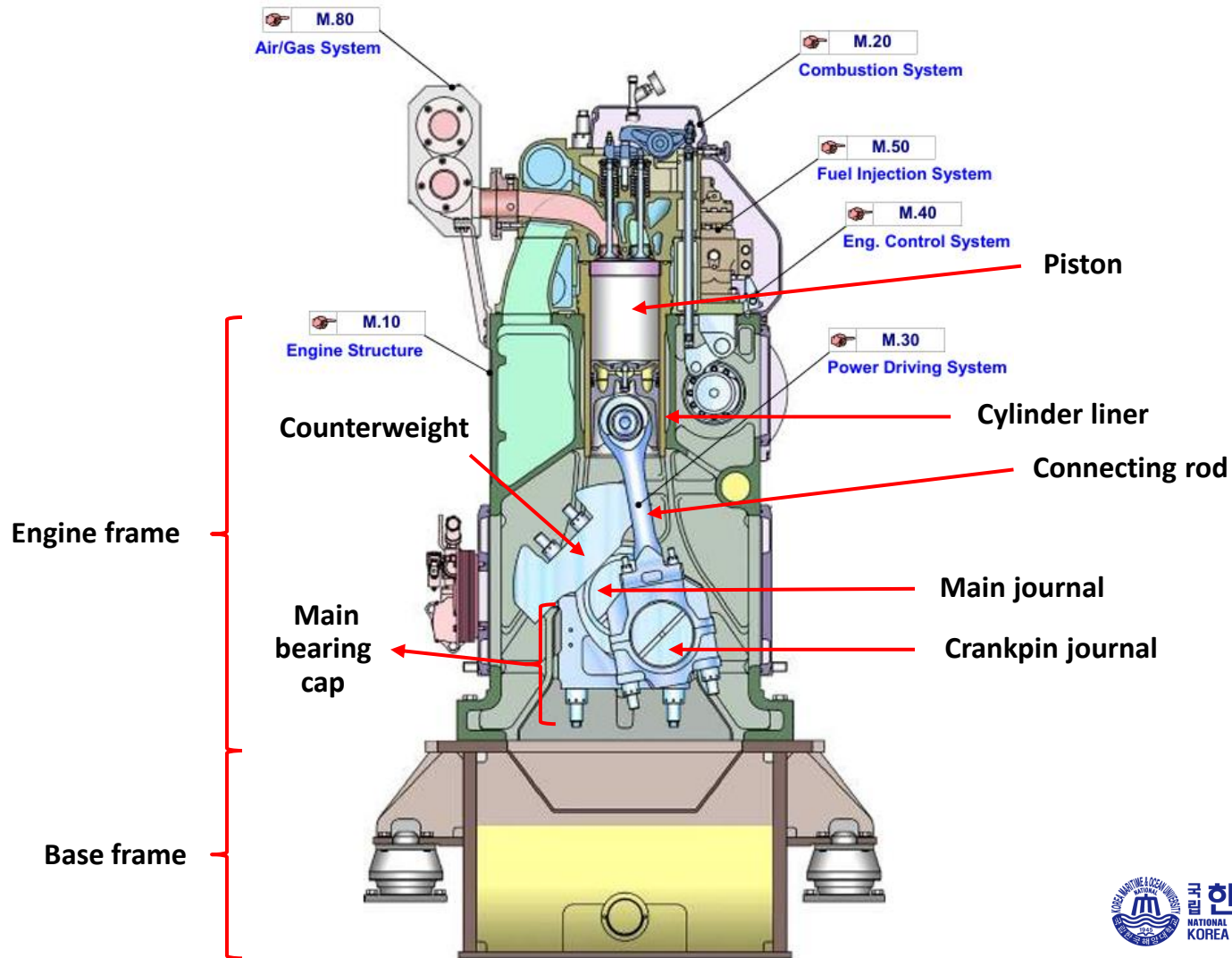
As the gear of the air starter is connected here and rotates, the crankshaft eventually rotates as well.



Crankshaft







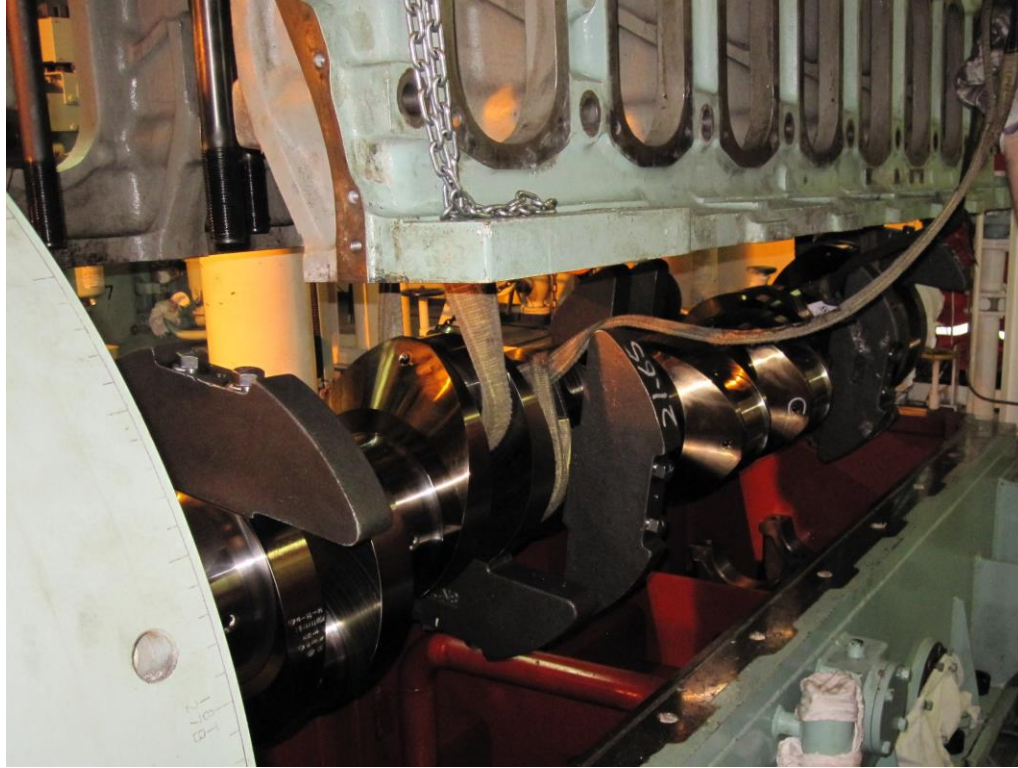
Crankshaft



Crankshaft



Crankshaft



Crankshaft deflection

Why crankshaft deflection is needed?

1. Crankshaft Alignment Check

The crankshaft is supported on multiple main bearings. Even slight misalignment of the bearings can cause shaft deflection. Measurements are taken to ensure the shaft remains straight and that the bearings provide even support.

2. Preventing Bearing Damage

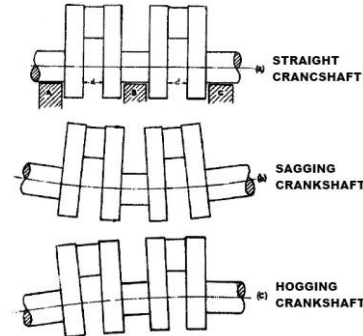
Misalignment can cause a concentrated load on a specific bearing, leading to bearing overheating, wear, and damage. Deflection measurements detect these problems early and prevent accidents.

3. Detecting Thermal and Structural Deformation

Thermal expansion during engine operation can cause the crankshaft to deform. Periodically measure the crankshaft to detect structural deformation.

4. Preventing Crankshaft Damage

Operating with excessive shaft deflection can lead to accumulated fatigue stress, which can lead to crankshaft cracking and damage. Deflection measurements are a key step in preventing damage.



Crankshaft deflection

1. Mount dial gauge between two crank webs of last cylinder nearest flywheel as shown in fig.1.

2. Rotate the crankshaft to measuring position 1 as shown in fig. 2.

3. Set the dial gauge to zero.

4. Rotate the crankshaft and read the values of dial gauge at measuring position 2 to 5, and then record these readings.

(Only the deviation from zero setting of dial gauge must be recorded.)

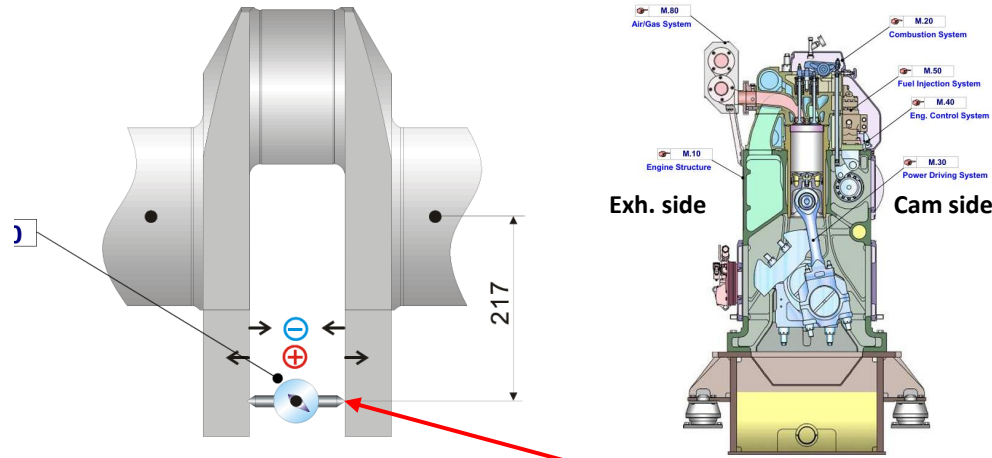


Fig.1 Installation of dial gauge (다이얼 게이지의 설치)

There is a groove in the crank web into which a dial gauge can be installed.

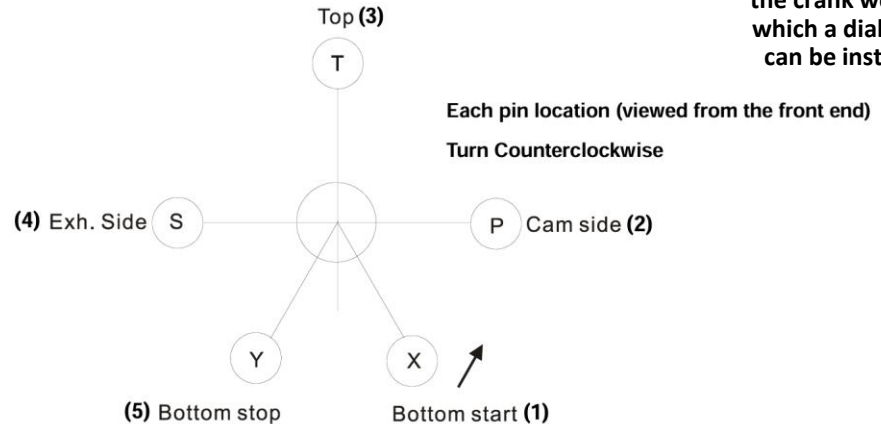


Fig.2 The measuring positions for deflection of crankshaft
(크랭크 샤프트의 치점량 측정 위치)

Crankshaft deflection

5. Evaluate these readings.

(1) A reading at measuring position 5 should be nearly the same as reading at measuring position 1. If larger differences, repeat steps 2 to 5.

(2) Vertical and horizontal deflections are as follows.

- Vertical deflection: $dv = T - (X+Y)/2$

- Horizontal deflection: $dh = P - S$

(If vertical and horizontal deflections exceed the permissible limits (+0.1 ~ -0.22 mm), realign the engine and the driven machinery.)

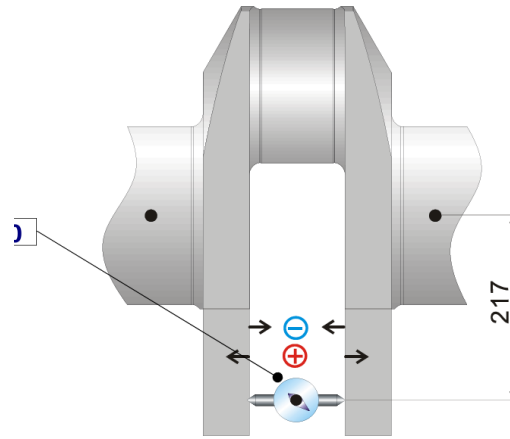


Fig.1 Installation of dial gauge (다이얼 게이지의 설치)

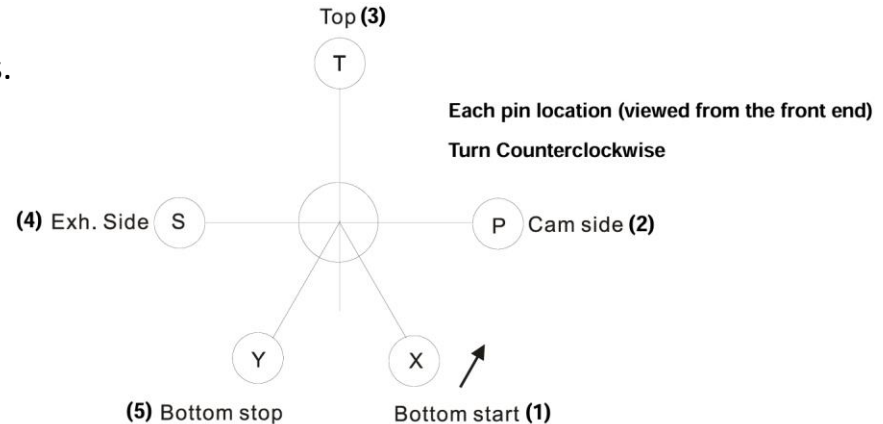
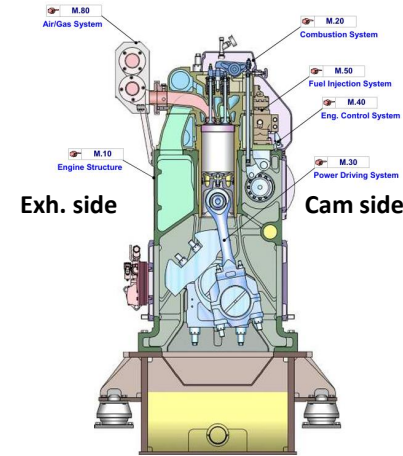


Fig.2 The measuring positions for deflection of crankshaft
(크랭크 샤프트의 치점량 측정 위치)

Crankshaft deflection

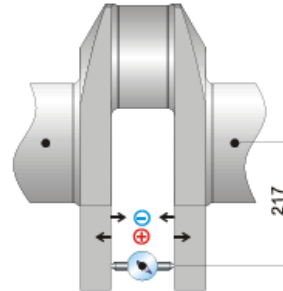
PERMISSIBLE LIMITS

(unit : 1 / 100 mm)

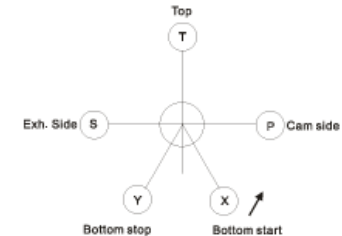
THE PERMISSIBLE DEFLECTION OF CRANKSHAFT		
Unless otherwise stated the values refer to cold engine.		
For the crank nearest the flywheel (Cold Condition)	New or recently overhauled engine, permissible	-26 ~ +12
	Operating engine, contact engine maker (check alignment immediately)	-28 ≥ or +17 ≤
For the other cranks (Cold Condition)	New or recently overhauled engine, permissible	-13 ~ +13
	Operating engine, check alignment immediately	-15 ≥ or +15 ≤

- * The values measured under hot condition are only for reference due to the following reason;
- If the values measured under cold condition satisfy the above mentioned specification, we guarantee any values measured under hot condition.

MEASURING POSITIONS AND POINTS



Gauge location on crank webs



Each pin location (viewed from the front end)
Turn Counterclockwise

** If the deflection value is over the above specifications, you should contact the engine maker immediately.

Crankshaft deflection

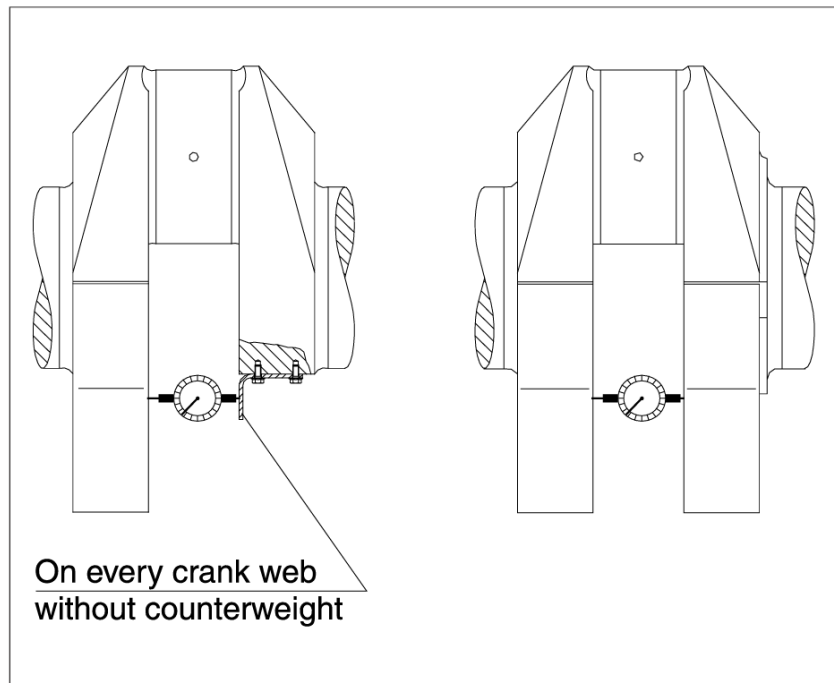


Fig. 1 Placing of dial gauge

Deflection of crankshaft in 1/100 mm. (0.01 mm).

Crank position		Cyl. No.					
		1	2	3	4	5	6
Bottom start	X	0	0	0	0	0	0
Left side	P	2	0	2	0	-1	2
Top	T	3	-2	4	5	-2	3
Right side	S	3	-2	2	0	0	1
Bottom stop	Y	2	-1	0	1	0	2

Fig. 2.

Bottom (0,5 x Y) = B	1	-0,5	0	-0,5	0	1
----------------------	---	------	---	------	---	---

Fig. 3.

Deflection from vertical misalignment							
top - bottom or T - B =	V	2	-1,5	4	4,5	-2	2

Fig. 4.

Deflection from horizontal misalignment							
Right side - left side or P - S =	H	-1	2	0	0	-1	1

Fig. 5.

Check on gauge readings	T + B = C	4	-2,5	4	5,5	-2	4
	P + S = D	5	-2	4	0	-1	3

Fig. 6.

C and D should be nearly the same, reading for cylinder 4 to be repeated.

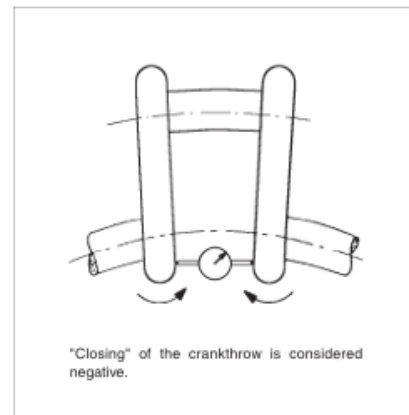


Fig. 7.

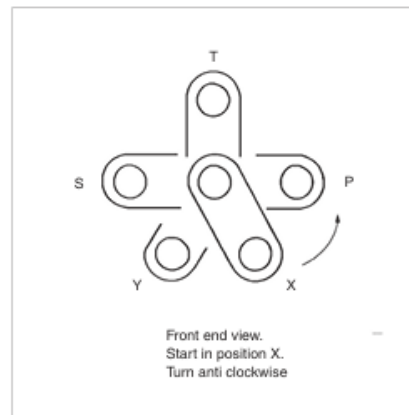


Fig. 8.

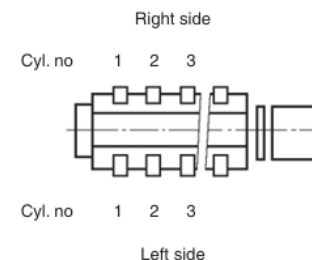
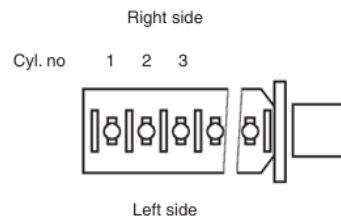
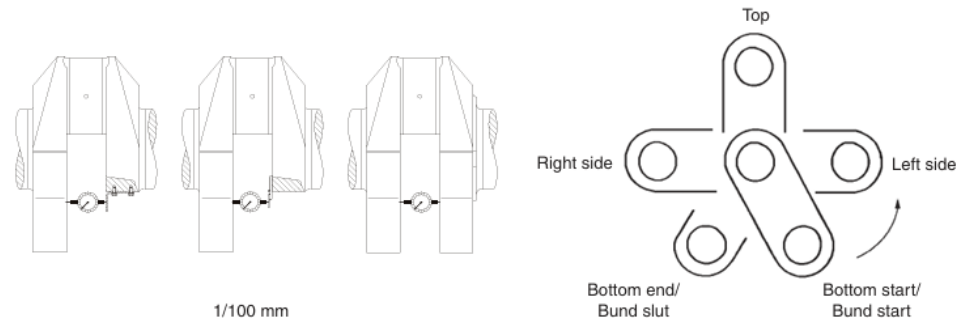
Measurement of Crank Throw Deflections by Means of Dial Indicator (Autolog)		
Crank throw deflection	=	Difference in dial indicator readings in two diametrically opposite crank throw positions, i.e. two positions displaced 180°.
Vertical deflection	=	Difference in top-bottom readings.
Horizontal deflection	=	Difference in side-side readings.

Vertical and Horizontal Deflections of Crank Throws		
Unless otherwise stated the values refer to cold engine.		
For new or realigned aggregate	Aim for	+ or - 4/100 mm
	Acceptable	+ or - 6/100 mm
For aggregate in service realignment is recommended if deflections exceed		+ or -10/100 mm

Vertical Deflection of Crank Throw at Flywheel	
Unless otherwise stated the values refer to cold engine.	
Rigid coupling between diesel engine and driven machine	Flexible coupling between diesel engine and driven machine
For new or realigned aggregate 0 to + 4/100 mm	For new or realigned aggregate Aim for - 7/100 mm Acceptable - 10/100 mm
For aggregate in service realignment is recommended if deflection measured on warm engine exceeds - 10/100 mm	For aggregate in service realignment recommended if deflection exceeds - 14/100 mm

Process/Proces		I.D. no.
Plant/Anlæg		Page of/Side af
Engine Type/Motortype	Engineer/Operator	Date/Dato

Instruction/Instruktion



Remarks/Bemærkninger

Component/Komponent	Type	I.D. no.
Process/Proces		Page of/Side af

Test place/ Condition	☐ Test bed/prøvehal	☐ Cold/Kold							
Teststed/ Tilstand	☐ On board/Om bord	☐ Warm/Varm							
	☐ Plant/Maskinhal								
Engine no.: Motornr.:									
Cyl. no.	1	2	3	4	5	6	7	8	9
Bottom	X	0	0	0	0	0	0	0	0
Left side	P								
Top	T								
Right side	S								
Bottom	Y								

Bottom (0.5xY)=B									
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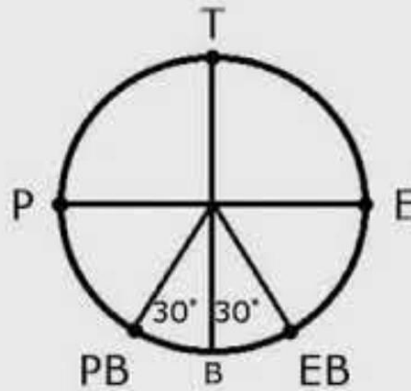
Deflection from vertical misalign- ment. Top - bottom or T - B = V									
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Deflection from horizontal mis- alignment. Left side - Right side or P - S = H									
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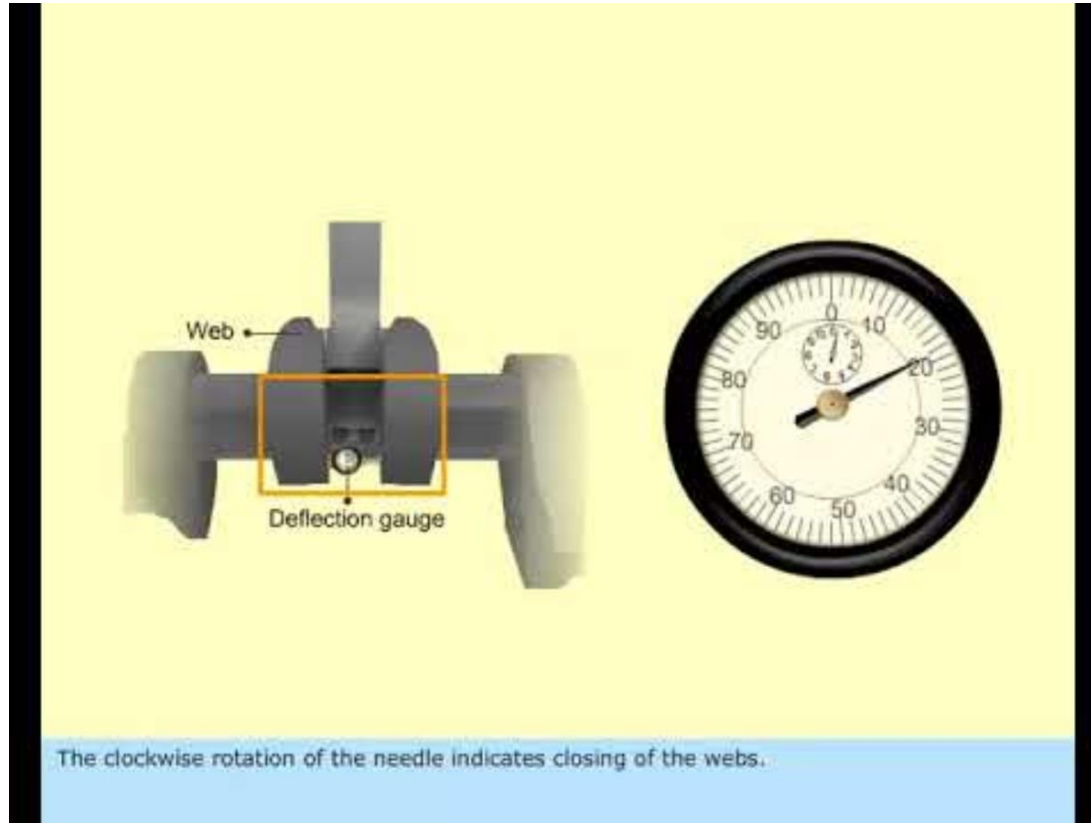
Check on gauge readings.									
T + B = C									
P + S = D									



Crankshaft deflection



Crankshaft deflection



Crankshaft deflection



Crankshaft deflection



References

- 1) <https://www.vcalc.com/wiki/engine/bore-stroke-ratio>
- 2) <https://www.businesskorea.co.kr/news/articleView.html?idxno=20316>
- 3) https://www.youtube.com/watch?v=D5_s7g8__xg
- 4) <https://chineseengine.en.made-in-china.com/product/pnxcysjHRUv/China-Hanshin-6EL32-Diesel-Engine-Spare-Parts-Piston-Ring-320mm.html>
- 5) <https://www.indiamart.com/proddetail/yanmar-s165-connecting-rod-22188597191.html>
- 6) <https://www.indiamart.com/proddetail/yanmar-s165-connecting-rod-22188597191.html>
- 7) <https://shipsparesindia.com/product/man-bw-l16-24-connecting-rod-bearing-standard/>
- 8) <https://shipsparesindia.com/product/man-bw-23-30-cr-bearing/>



References

- 9) <https://shipsparesindia.com/product/man-bw-23-30-connecting-rod/>
- 10) https://www.smedegaarden.net/shop/engine_engine_parts/liner/man_b_w_alpha_liner_for_23_30
- 11) <https://hfoplant.blogspot.com/2011/10/cylinder-liner-installtion-process.html>
- 12) <https://www.industrialmarinepower.com/hyundaihimsen-8h2132-crankshaft-newunused-16106/>
- 13) <https://muckymariners.com/app/meo-class-4-subjects/function-4b-meol/engines/7-cranckshaft-crosshead/cranckshaft-crosshead-2-7-crankshaft-deflection/>
- 14) <https://www.goltens.com/resources/himsen-6h25-33-crankshaft-replacement-singapore/>
- 15) <https://www.oktotr.com/aux-engine-crankshaft-replacement-during-sailing>
- 16) <https://www.slideserve.com/carmelita/lubrication-features-of-a-large-diesel-engine>
- 17) https://www.researchgate.net/figure/Schematic-representation-of-the-piston-group-and-liner-assembly-of-the-diesel-off-highway_fig1_351112570
- 18) <https://farmmachinerydigest.com/blowing-in-the-wind-blow-by-on-large-diesel-engines/>



References

- 19) <https://www.youtube.com/watch?v=awIHn-8H6WM>
- 20) <https://www.youtube.com/watch?v=pSJlHrdR4mk>
- 21) <https://www.youtube.com/watch?v=I9wMYxNar0k>
- 22) <https://www.youtube.com/watch?v=AeSxK0Sn2Qg>
- 23) https://www.researchgate.net/figure/Procedure-for-liquid-penetrant-test-10_fig1_355889692
- 24) <https://www.nationalboard.org/index.aspx?pageID=164&ID=374>
- 25) <https://www.youtube.com/watch?v=tWIP22-Xcxs>
- 26) <https://shop.idealprec.com/products/mitutoyo-339-304-digimatic-tubular-inside-micrometer-8-80-203-1524-range-0001-0-001mm-resolution>



References

27) <https://www.youtube.com/shorts/3U6F1T2azKg>

28) <https://www.youtube.com/shorts/O3NTbO4RhAM>

